

EC4060 – COMPUTER AND DATA NETWORK
INDEPENDENT LEARNING AND IMPLEMENTATION

ASSIGNMENT -I
SIMULATING NETWORK INFRASTRUCTURE DESIGN
FOR A MULTI-BRANCH FACILITIES

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GROUP: EG08

SEMESTER: 04

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Objective

- ✓ Apply the principles of network design to create and simulate a functional network infrastructure for an institution with multiple branches.

Scenario

- ✓ The Engineering Faculty consists of 5 academic departments (Civil, Mechanical, EEE, Computer, and Interdisciplinary Studies) and 1 Administration Section, requiring a scalable and secure network.

Task

- ✓ The student is tasked with designing and simulating this network while ensuring:
 - Unique subnet allocation for each section.
 - Identification of subnet information, including subnet mask, usable host range, and broadcast address.
 - Scalability to accommodate at least 30% future growth in each section.

Deliverables

- ✓ A Report and Simulation Video
 1. Subnet Calculation Table
 2. Network Design Diagram
 3. Simulation File
 4. Configuration Scripts
 5. Test Report

Categorization of Devices

- ✓ Separate devices for staff and students within each department.

Common Devices

- ✓ Include department-specific printers and shared devices accessible only by staff.

Central CCTV System

- ✓ A unified subnet for CCTV cameras covering all departments.

End Devices

Department	Common Computers	Staff Computers	Printers	Other Devices
Computer Eng	250	50	2	Min 25
EE Eng	150	50	2	Min 15
Civil Eng	75	25	2	Min 5
Mech Eng	75	25	2	Min 10
IDS	15	25	2	Min 5
Administration	0	25	5	0

No of Devices with at least 30% future growth in each section

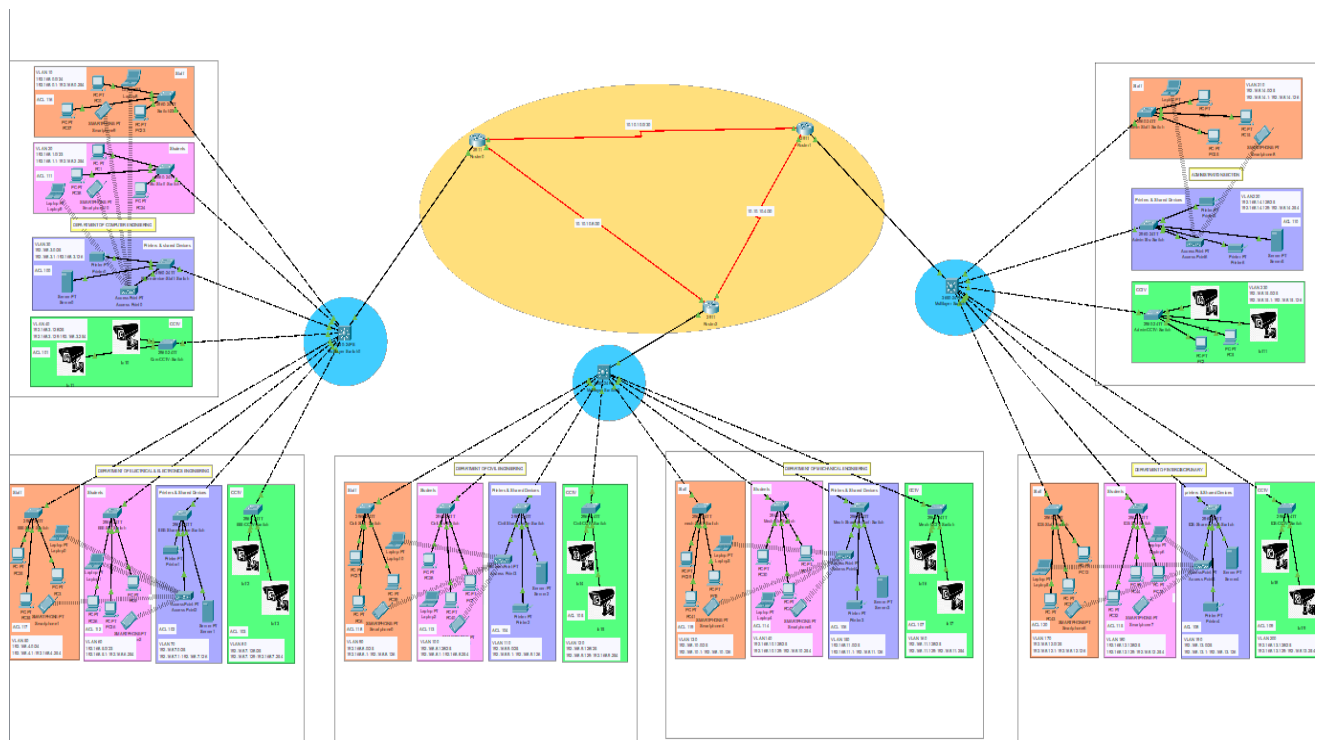
Devices	Departments					Admin
	Computer	EEE	Civil	Mechanical	IDS	
Common	325	195	98	98	20	-
Staff	65	65	33	33	33	33
Printers	3	3	3	3	3	7
Other	33	20	7	13	7	-
Total	426	283	141	147	63	40

Subnet Calculation Table

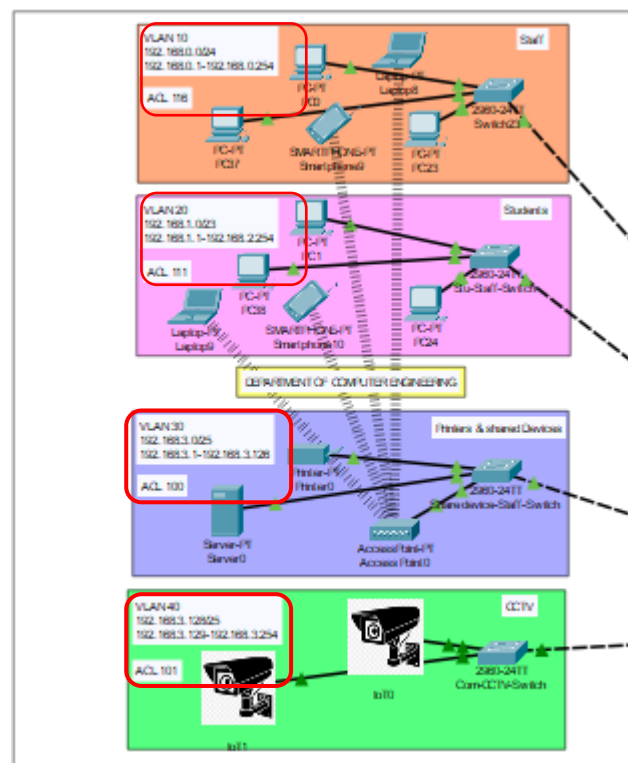
Department	Category	VLAN	Subnet Address
Computer	Staff	10	192.168.0.0/24
	Student	20	192.168.0.128/23
	Shared	30	192.168.2.0/25
	CCTV	40	192.168.3.128/25
EEE	Staff	50	192.168.4.0/24
	Student	60	192.168.5.0/23
	Shared	70	192.168.7.0/25
	CCTV	80	192.168.7.128/25
Civil	Staff	90	192.168.8.0/25
	Student	100	192.168.8.128/25
	Shared	110	192.168.9.0/25
	CCTV	120	192.168.9.0/25
Mechanical	Staff	130	192.168.10.0/25
	Student	140	192.168.10.128/25
	Shared	150	192.168.11.0/25
	CCTV	160	192.168.11.128/25
IDS	Staff	170	192.168.12.0/25
	Student	180	192.168.12.128/25
	Shared	190	192.168.13.0/25
	CCTV	200	192.168.13.128/25
Admin	Student	210	192.168.14.0/25
	Shared	220	192.168.14.128/25
	CCTV	230	192.168.15.0/25

Department	Network Address	Broad Cast Address	Subnet Mask	IP Range
Computer	192.168.0.0	192.168.0.255	255.255.255.0	192.168.0.1-192.168.0.254
	192.168.0.128	192.168.2.255	255.255.254.0	192.168.1.1-192.168.2.254
	192.168.2.0	192.168.3.127	255.255.255.128	192.168.3.1-192.168.3.126
	192.168.3.128	192.168.3.255	255.255.255.128	192.168.3.129-192.168.3.254
EEE	192.168.4.0	192.168.4.255	255.255.255.0	192.168.4.1-192.168.4.254
	192.168.5.0	192.168.6.255	255.255.254.0	192.168.5.1-192.168.6.254
	192.168.7.0	192.168.7.127	255.255.255.128	192.168.7.1-192.168.7.126
	192.168.7.128	192.168.7.255	255.255.255.128	192.168.7.129-192.168.7.254
Civil	192.168.8.0	192.168.8.127	255.255.255.128	192.168.8.1-192.168.8.126
	192.168.8.128	192.168.8.255	255.255.255.128	192.168.8.129-192.168.8.254
	192.168.9.0	192.168.9.127	255.255.255.128	192.168.9.1-192.168.9.126
	192.168.9.128	192.168.9.255	255.255.255.128	192.168.9.129-192.168.9.254
Mechanical	192.168.10.0	192.168.10.127	255.255.255.128	192.168.10.1-192.168.10.126
	192.168.10.128	192.168.10.255	255.255.255.128	192.168.10.129-192.168.10.254
	192.168.11.0	192.168.11.127	255.255.255.128	192.168.11.1-192.168.11.126
	192.168.11.128	192.168.11.255	255.255.255.128	192.168.11.129-192.168.11.254
IDS	192.168.12.0	192.168.12.127	255.255.255.128	192.168.12.1-192.168.12.126
	192.168.12.128	192.168.12.255	255.255.255.128	192.168.12.129-192.168.12.254
	192.168.13.0	192.168.13.127	255.255.255.128	192.168.13.1-192.168.13.126
	192.168.13.128	192.168.13.255	255.255.255.128	192.168.13.129-192.168.13.254
Admin	192.168.14.0	192.168.14.127	255.255.255.128	192.168.14.1-192.168.14.126
	192.168.14.128	192.168.14.255	255.255.255.128	192.168.14.129-192.168.14.254
	192.168.15.0	192.168.15.127	255.255.255.128	192.168.15.1-192.168.15.126

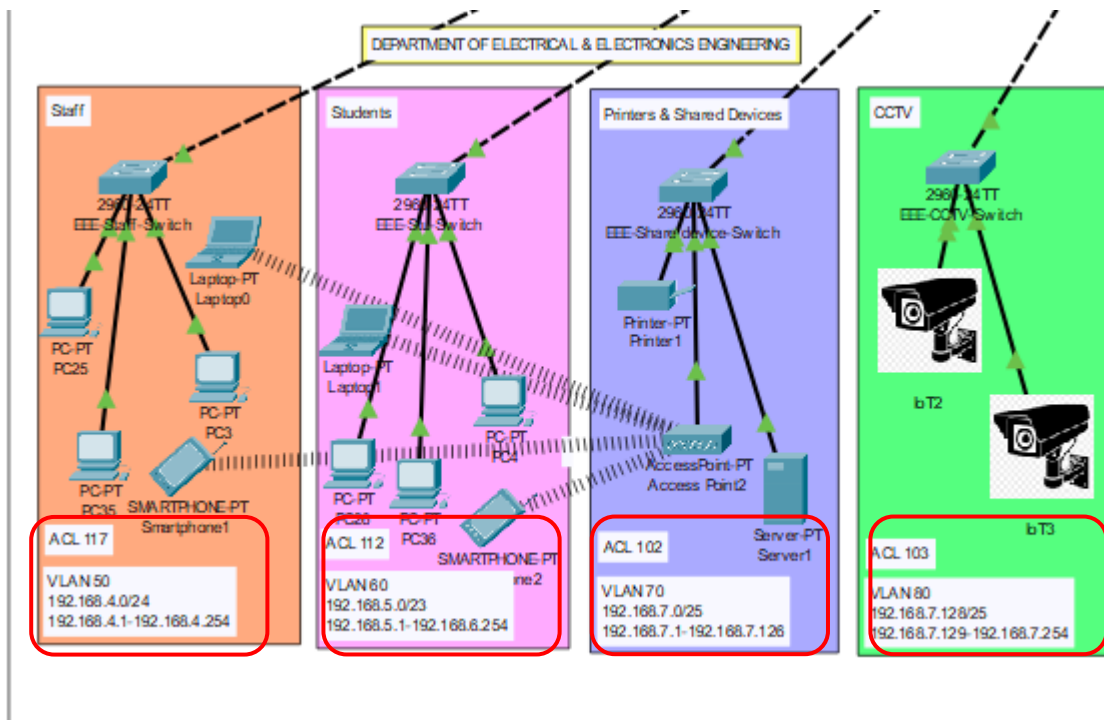
Network Design Diagram(Topology Diagram)



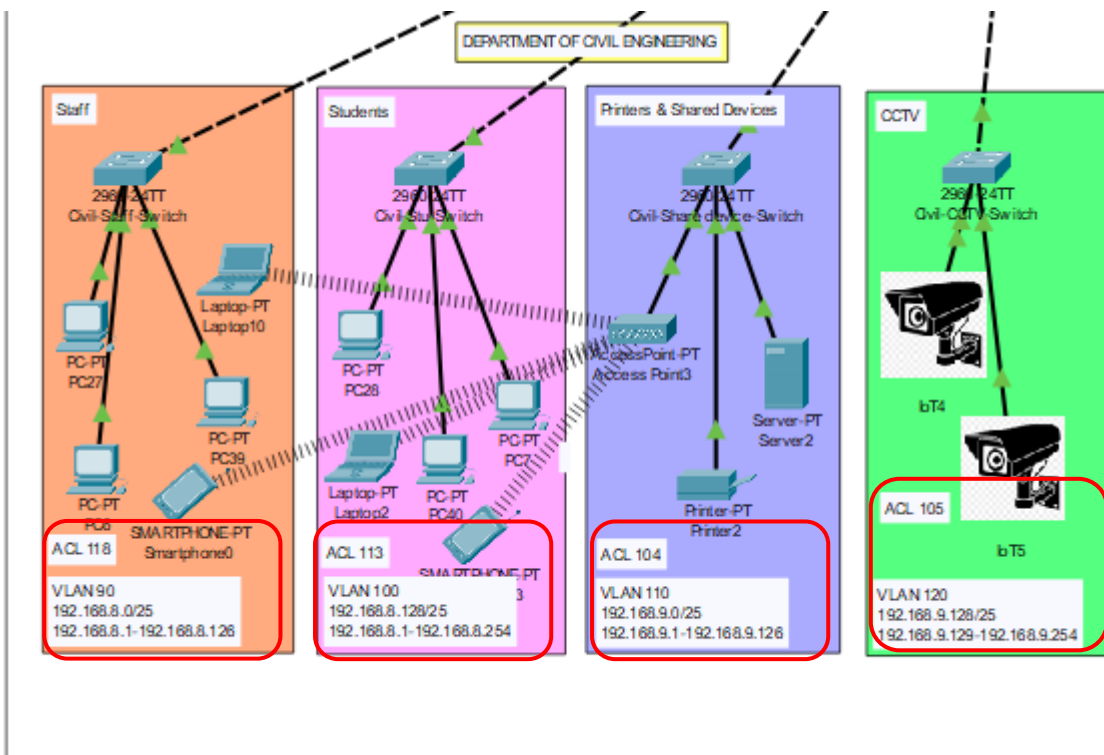
I. Department of Computer Engineering



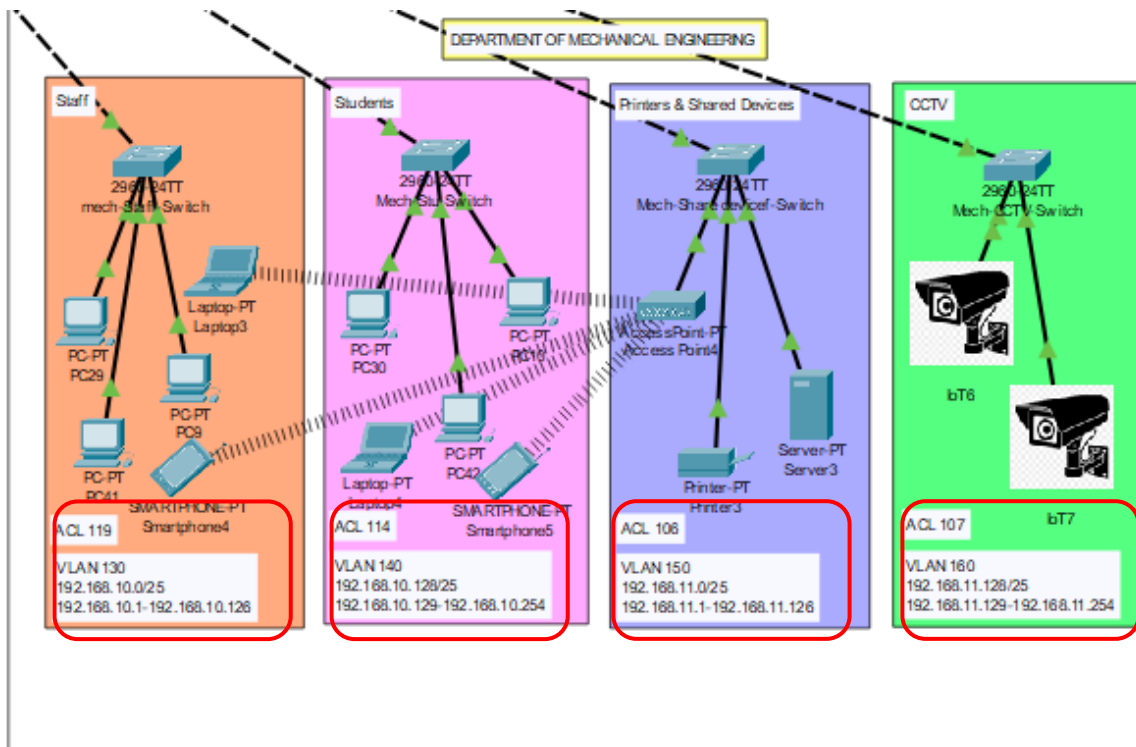
II. Department of Electrical & Electronics Engineering



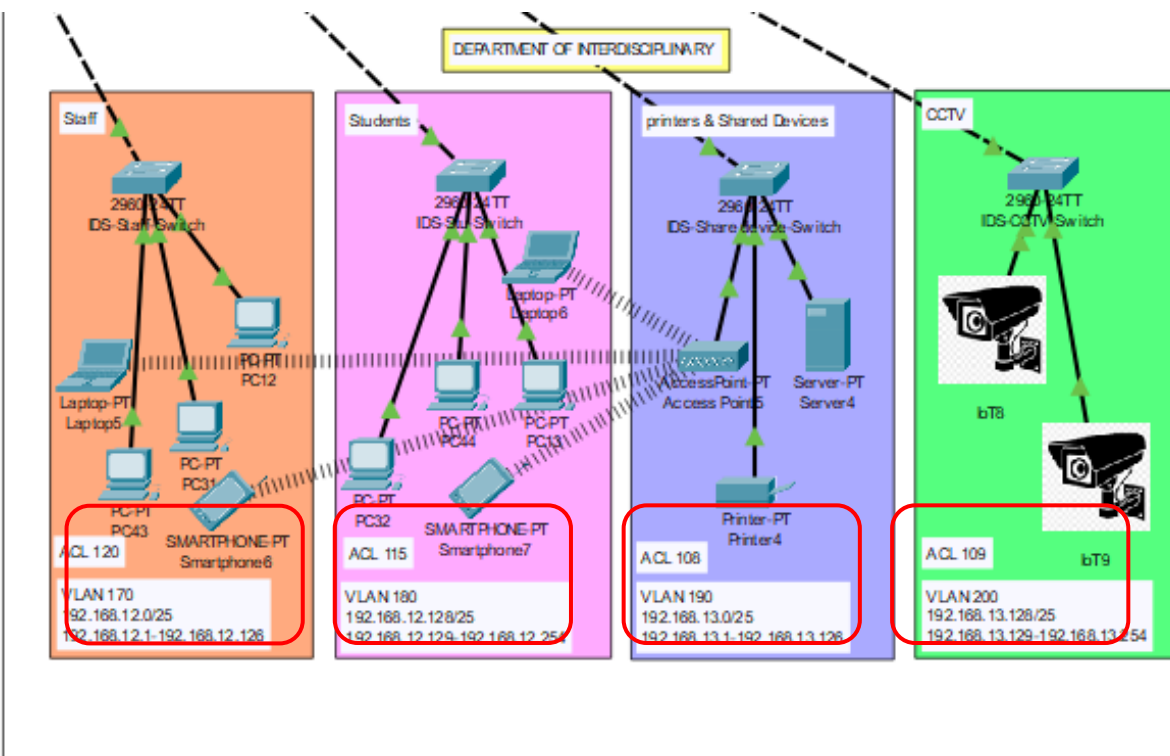
III. Department of Civil Engineering



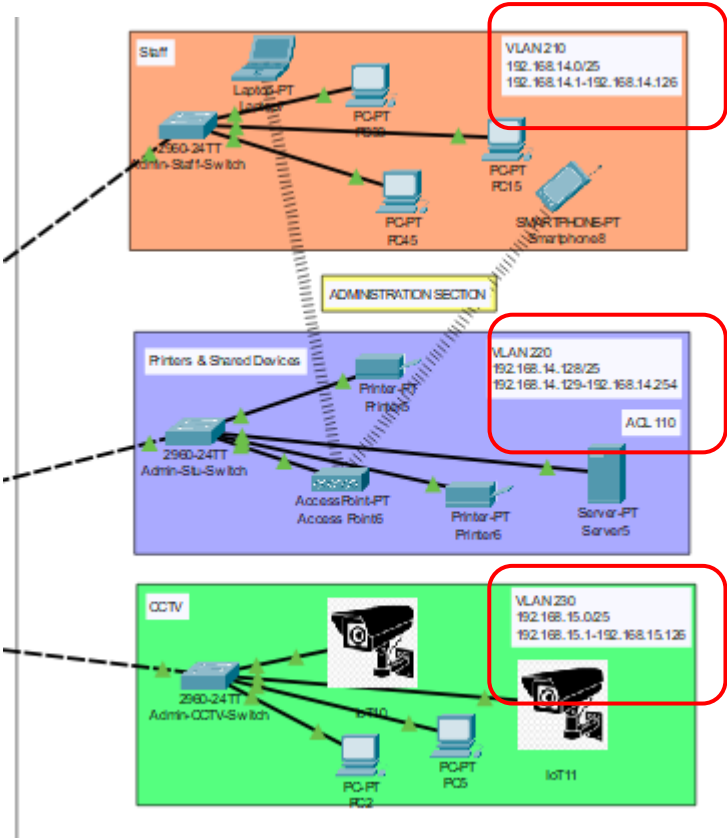
IV. Department of Mechanical Engineering



V. Department of Interdisciplinary

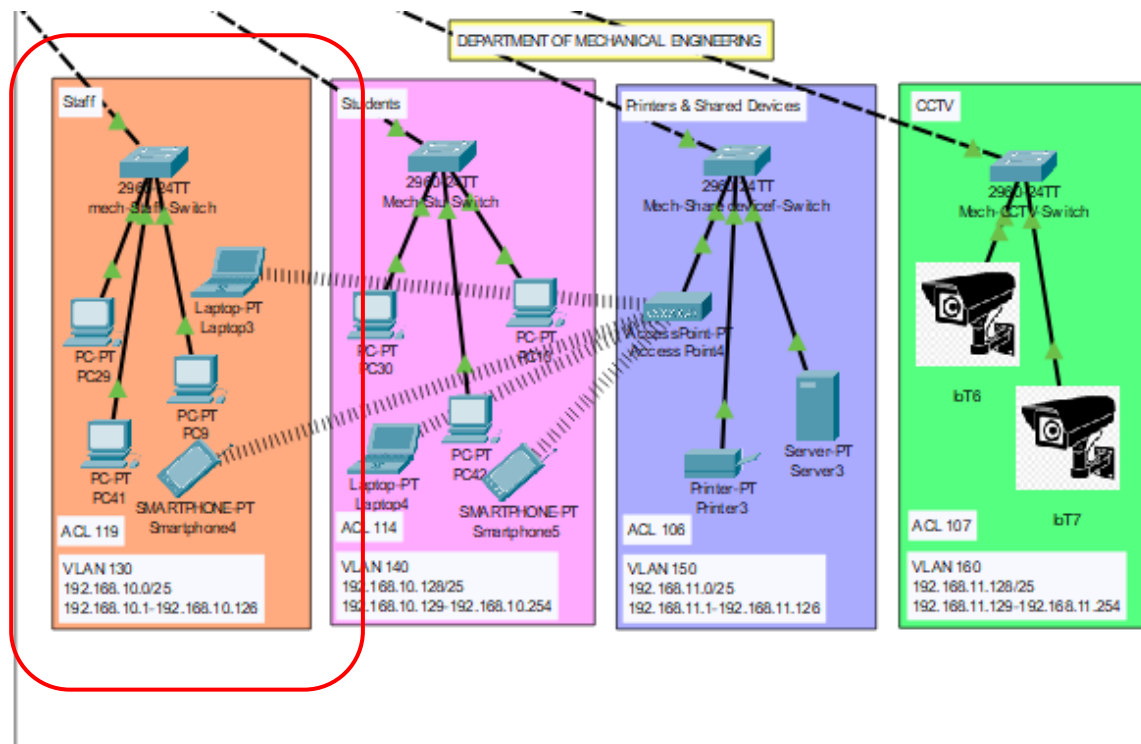
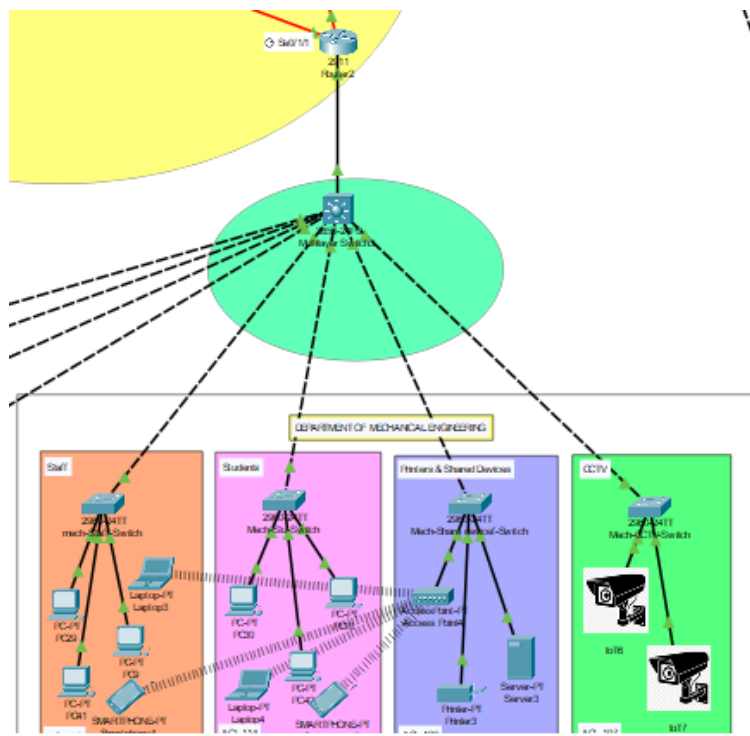


VI. Administration Section



Configuration Scripts

- ✓ Configuration scripts procedures using Department of Mechanical Engineering(Staff)



I. Configuration of Switch



The screenshot shows the CLI window for 'mech-Staff-Switch'. The 'CLI' tab is selected. The window displays the following text:

```
IOS Command Line Interface

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2,
changed state to up

%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1,
changed state to up

%LINK-5-CHANGED: Interface FastEthernet0/3, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/3,
changed state to up

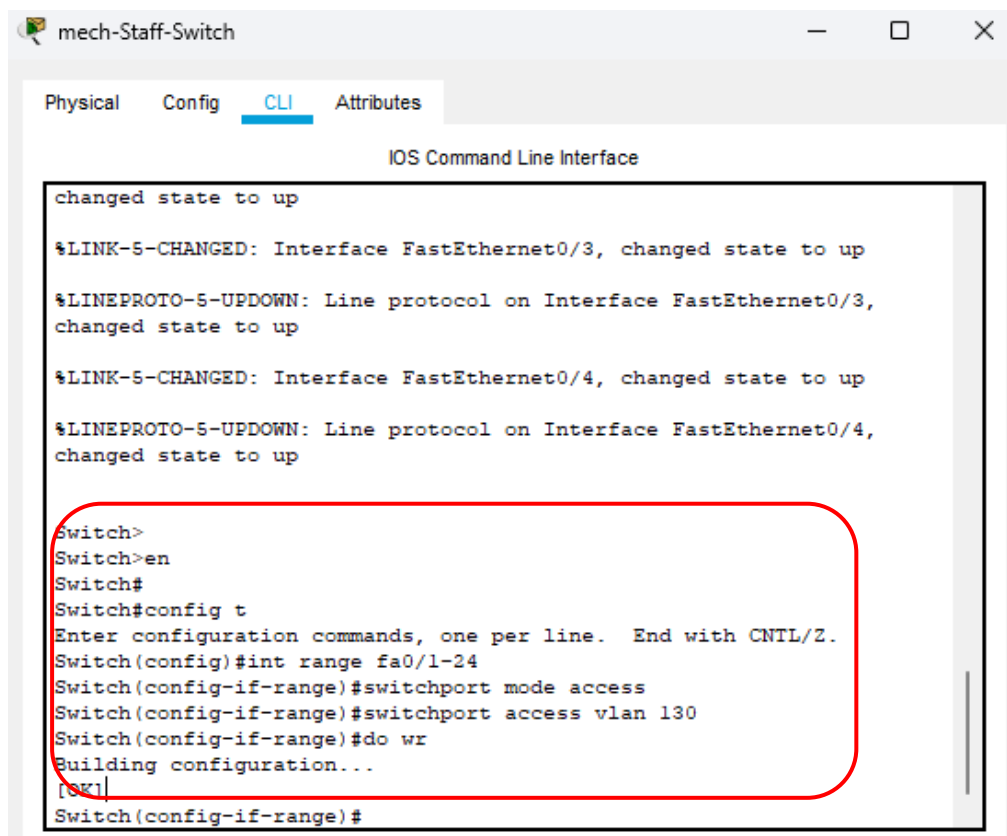
%LINK-5-CHANGED: Interface FastEthernet0/4, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/4,
changed state to up

Switch>
Switch>en
Switch#
Switch#config t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#
```

The bottom portion of the CLI window, starting from 'Switch>' and ending at 'Switch(config)#', is highlighted with a red rounded rectangle.

II. VLANs accessed by the Switch are initiated



The screenshot shows the CLI window for 'mech-Staff-Switch'. The 'CLI' tab is selected. The window displays the following text:

```
IOS Command Line Interface

changed state to up

%LINK-5-CHANGED: Interface FastEthernet0/3, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/3,
changed state to up

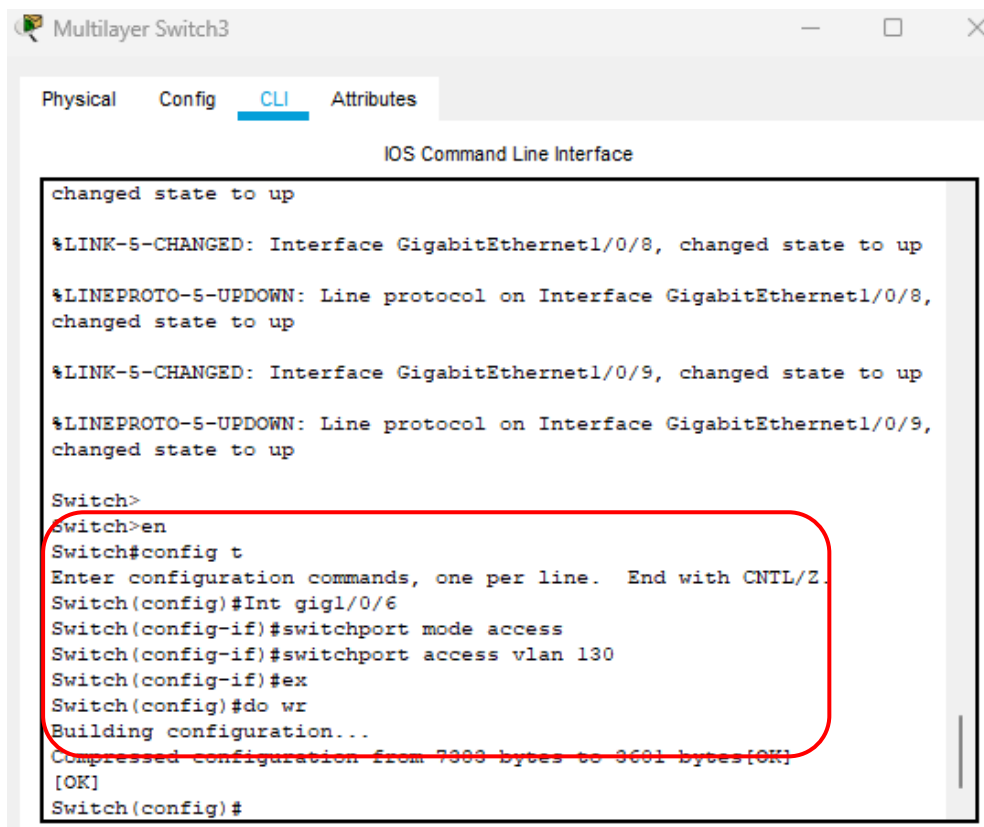
%LINK-5-CHANGED: Interface FastEthernet0/4, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/4,
changed state to up

Switch>
Switch>en
Switch#
Switch#config t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#int range fa0/1-24
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#switchport access vlan 130
Switch(config-if-range)#do wr
Building configuration...
[OK]
Switch(config-if-range)#
```

The bottom portion of the CLI window, starting from 'Switch>' and ending at 'Switch(config-if-range)#', is highlighted with a red rounded rectangle.

III. Configuration of the multilayer Switch and VLAN access



The screenshot shows the CLI of a Multilayer Switch3. The 'CLI' tab is selected. The output shows the state of interfaces GigabitEthernet1/0/8 and 1/0/9 changing to up. Then, the user enters configuration mode and configures interface GigabitEthernet1/0/6 as an access port in VLAN 130. The configuration is saved, compressing from 7383 bytes to 3601 bytes. The configuration commands are highlighted with a red box.

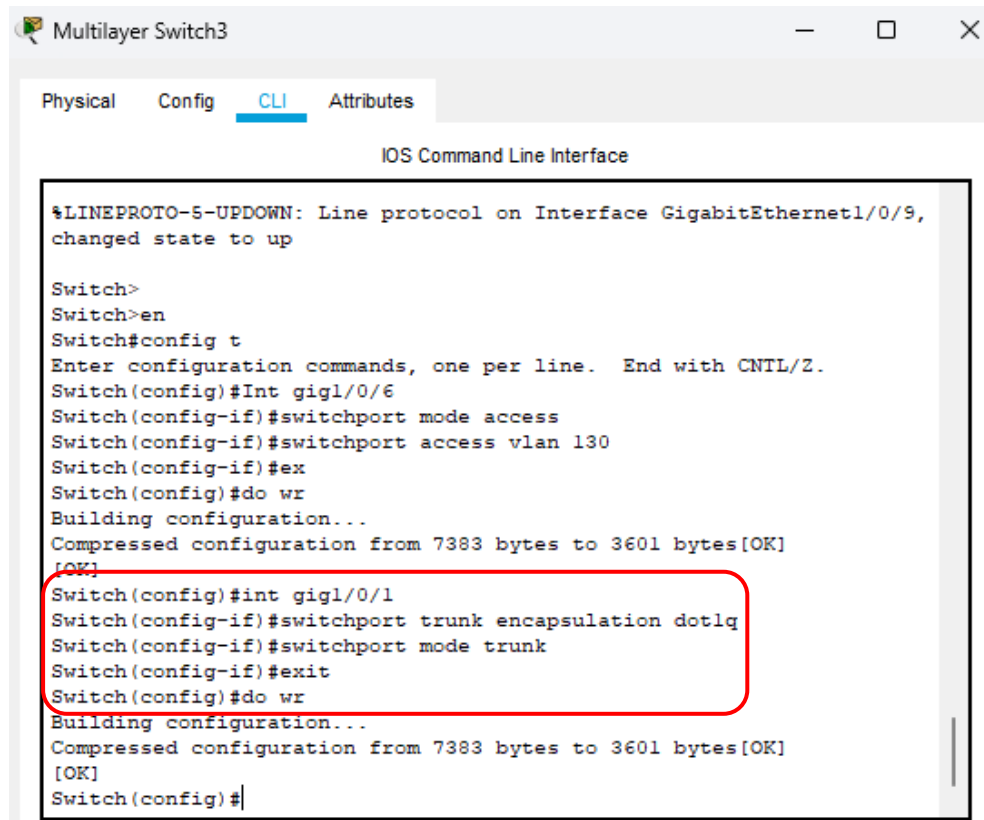
```
Multilayer Switch3
Physical Config CLI Attributes

IOS Command Line Interface

changed state to up
%LINK-5-CHANGED: Interface GigabitEthernet1/0/8, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet1/0/8,
changed state to up
%LINK-5-CHANGED: Interface GigabitEthernet1/0/9, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet1/0/9,
changed state to up

Switch>
Switch>en
Switch#config t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#Int gig1/0/6
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 130
Switch(config-if)#ex
Switch(config)#do wr
Building configuration...
Compressed configuration from 7383 bytes to 3601 bytes[OK]
[OK]
Switch(config)#
```

IV. Configure the trunk interface in multilayer switch



The screenshot shows the CLI of a Multilayer Switch3. The 'CLI' tab is selected. The output shows the state of interface GigabitEthernet1/0/9 changing to up. Then, the user enters configuration mode and configures interface GigabitEthernet1/0/1 as a trunk port with dot1q encapsulation. The configuration is saved, compressing from 7383 bytes to 3601 bytes. The configuration commands are highlighted with a red box.

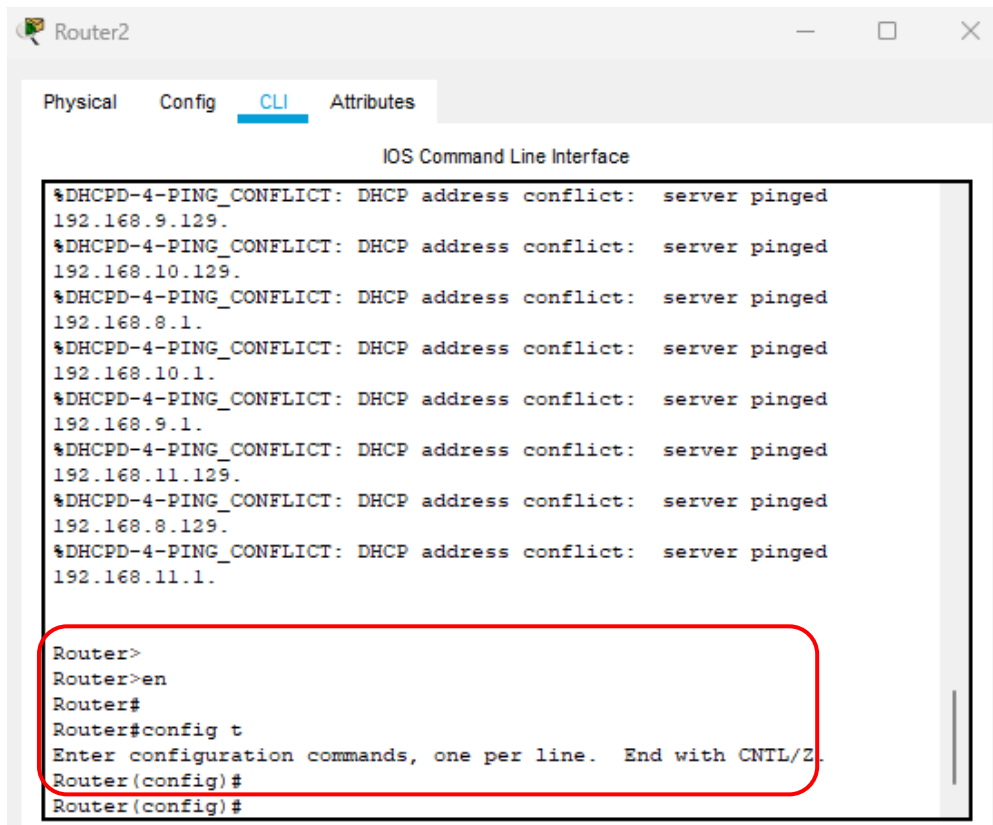
```
Multilayer Switch3
Physical Config CLI Attributes

IOS Command Line Interface

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet1/0/9,
changed state to up

Switch>
Switch>en
Switch#config t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#Int gig1/0/6
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 130
Switch(config-if)#ex
Switch(config)#do wr
Building configuration...
Compressed configuration from 7383 bytes to 3601 bytes[OK]
[OK]
Switch(config)#int gig1/0/1
Switch(config-if)#switchport trunk encapsulation dot1q
Switch(config-if)#switchport mode trunk
Switch(config-if)#exit
Switch(config)#do wr
Building configuration...
Compressed configuration from 7383 bytes to 3601 bytes[OK]
[OK]
Switch(config)#
```

V. Configuration of Router



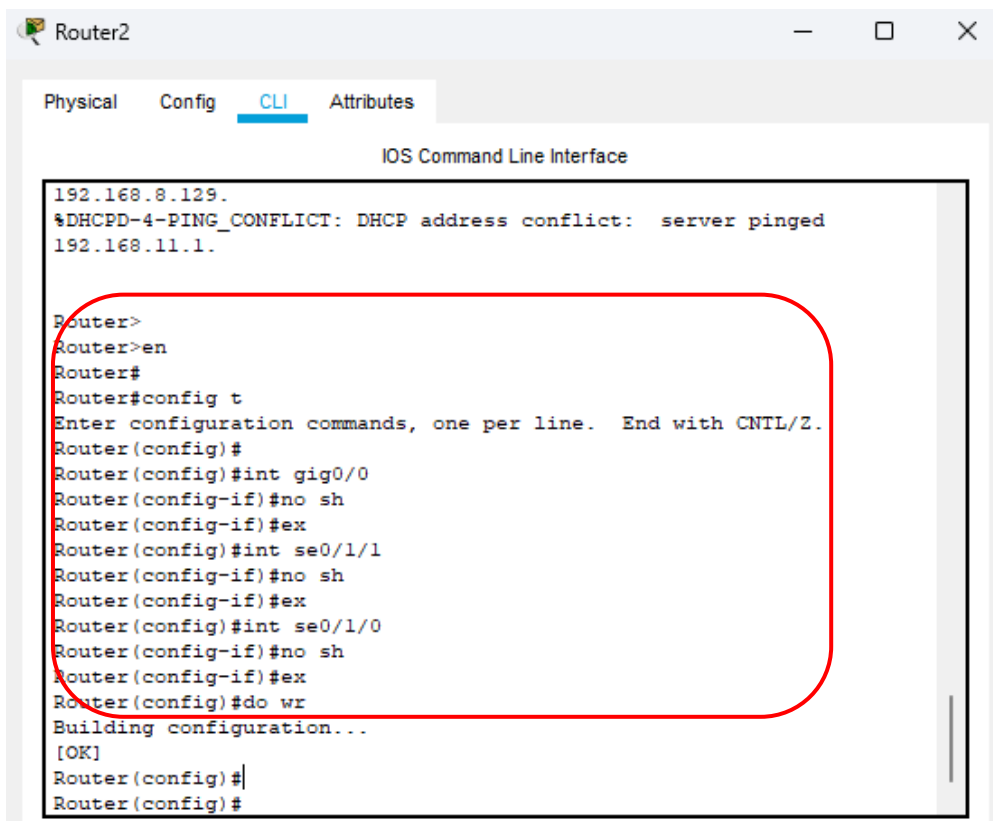
The screenshot shows a window titled "Router2" with tabs for "Physical", "Config", "CLI", and "Attributes". The "CLI" tab is selected, displaying the "IOS Command Line Interface". The interface shows a series of DHCP address conflict messages from a server, each followed by a pinged IP address. Below these messages, the user enters the command "en" to enter configuration mode, and then "config t" to enter global configuration mode. The prompt changes from "Router>" to "Router#" and then to "Router(config)#".

```
Router2
Physical Config CLI Attributes
IOS Command Line Interface

%DHCPD-4-PING_CONFLICT: DHCP address conflict: server pinged
192.168.9.129.
%DHCPD-4-PING_CONFLICT: DHCP address conflict: server pinged
192.168.10.129.
%DHCPD-4-PING_CONFLICT: DHCP address conflict: server pinged
192.168.8.1.
%DHCPD-4-PING_CONFLICT: DHCP address conflict: server pinged
192.168.10.1.
%DHCPD-4-PING_CONFLICT: DHCP address conflict: server pinged
192.168.9.1.
%DHCPD-4-PING_CONFLICT: DHCP address conflict: server pinged
192.168.11.129.
%DHCPD-4-PING_CONFLICT: DHCP address conflict: server pinged
192.168.8.129.
%DHCPD-4-PING_CONFLICT: DHCP address conflict: server pinged
192.168.11.1.

Router>
Router>en
Router#
Router#config t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#
Router(config)#
```

VI. Configure the router interface



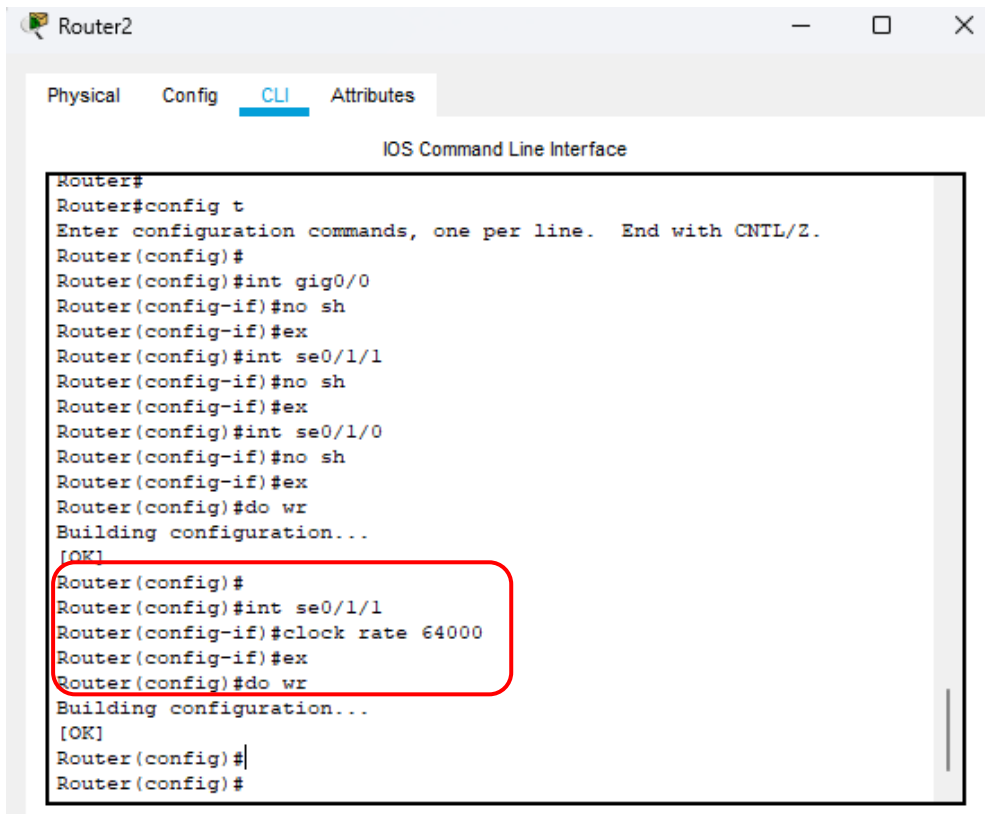
The screenshot shows the same "Router2" window with the "CLI" tab selected. The "IOS Command Line Interface" displays the continuation of the configuration process. The user enters "en" to enter configuration mode, then "config t" to enter global configuration mode. The prompt changes from "Router>" to "Router#" and then to "Router(config)#". The user then enters "int gig0/0" to enter interface configuration mode for GigabitEthernet 0/0. The prompt changes to "Router(config-if)#". The user enters "no sh" to disable shutdown, then "ex" to exit interface configuration mode. The prompt returns to "Router(config)#". The user then enters "int se0/1/1" to enter interface configuration mode for Serial 0/1/1. The prompt changes to "Router(config-if)#". The user enters "no sh" to disable shutdown, then "ex" to exit interface configuration mode. The prompt returns to "Router(config)#". The user then enters "int se0/1/0" to enter interface configuration mode for Serial 0/1/0. The prompt changes to "Router(config-if)#". The user enters "no sh" to disable shutdown, then "ex" to exit interface configuration mode. The prompt returns to "Router(config)#". The user then enters "do wr" to save the configuration. The prompt changes to "Router(config)#". The user then enters "do wr" to save the configuration. The prompt changes to "Router(config)#".

```
Router2
Physical Config CLI Attributes
IOS Command Line Interface

192.168.8.129.
%DHCPD-4-PING_CONFLICT: DHCP address conflict: server pinged
192.168.11.1.

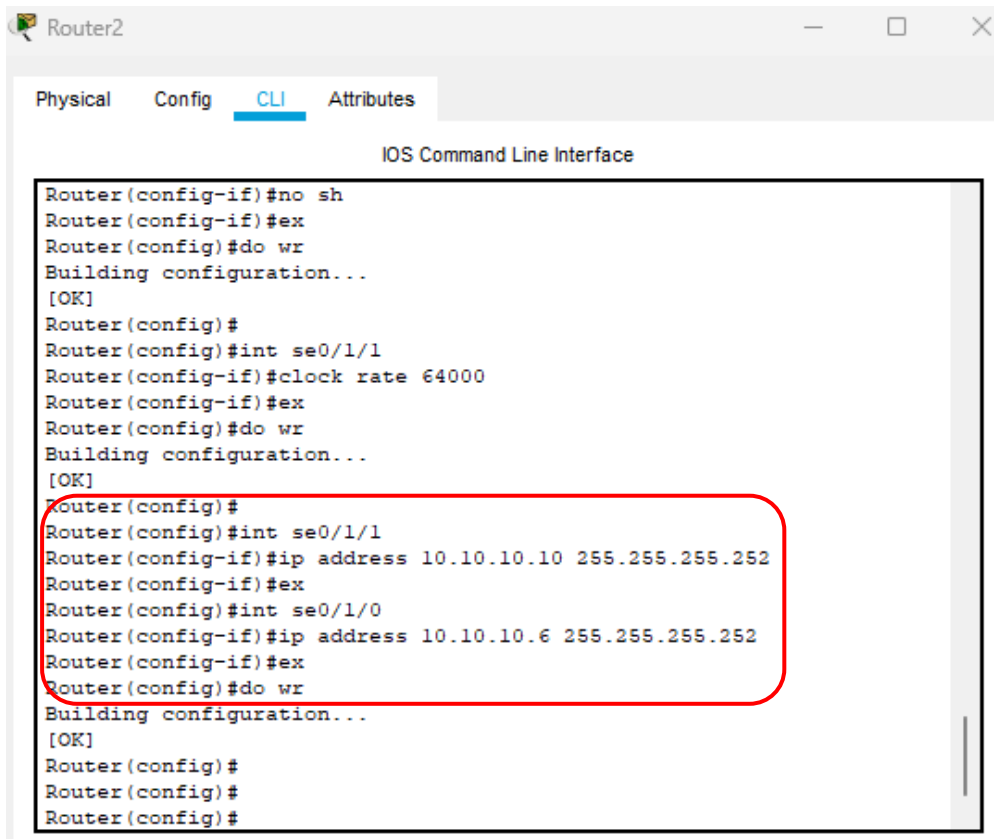
Router>
Router>en
Router#
Router#config t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#
Router(config)#int gig0/0
Router(config-if)#no sh
Router(config-if)#ex
Router(config)#int se0/1/1
Router(config-if)#no sh
Router(config-if)#ex
Router(config)#int se0/1/0
Router(config-if)#no sh
Router(config-if)#ex
Router(config)#do wr
Building configuration...
[OK]
Router(config)#
Router(config)#
```

VII. Set the clock rate to the Serial DCE interface of router



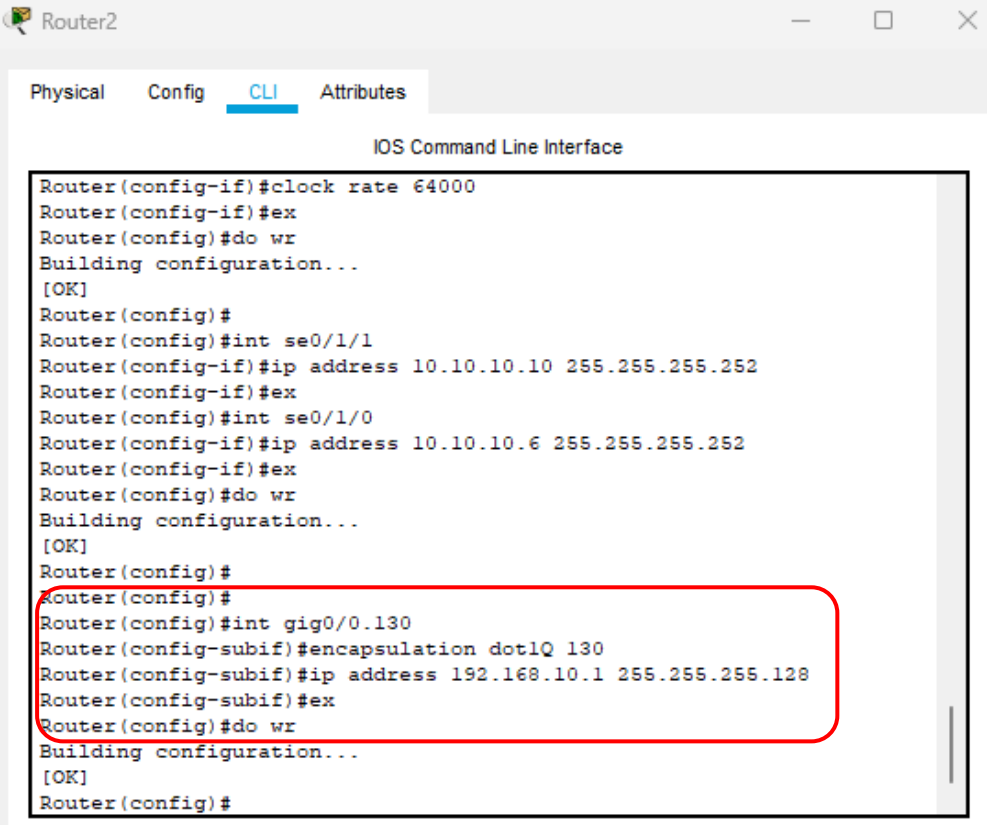
```
Router2
Physical Config CLI Attributes
IOS Command Line Interface
Router#
Router#config t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#
Router(config)#int gig0/0
Router(config-if)#no sh
Router(config-if)#ex
Router(config)#int se0/1/1
Router(config-if)#no sh
Router(config-if)#ex
Router(config)#int se0/1/0
Router(config-if)#no sh
Router(config-if)#ex
Router(config)#do wr
Building configuration...
[OK]
Router(config)#
Router(config)#int se0/1/1
Router(config-if)#clock rate 64000
Router(config-if)#ex
Router(config)#do wr
Building configuration...
[OK]
Router(config)#
Router(config)#
```

VIII. Set the IP for serial DCE interface of router



```
Router2
Physical Config CLI Attributes
IOS Command Line Interface
Router(config-if)#no sh
Router(config-if)#ex
Router(config)#do wr
Building configuration...
[OK]
Router(config)#
Router(config)#int se0/1/1
Router(config-if)#clock rate 64000
Router(config-if)#ex
Router(config)#do wr
Building configuration...
[OK]
Router(config)#
Router(config)#int se0/1/1
Router(config-if)#ip address 10.10.10.10 255.255.255.252
Router(config-if)#ex
Router(config)#int se0/1/0
Router(config-if)#ip address 10.10.10.6 255.255.255.252
Router(config-if)#ex
Router(config)#do wr
Building configuration...
[OK]
Router(config)#
Router(config)#
Router(config)#
```

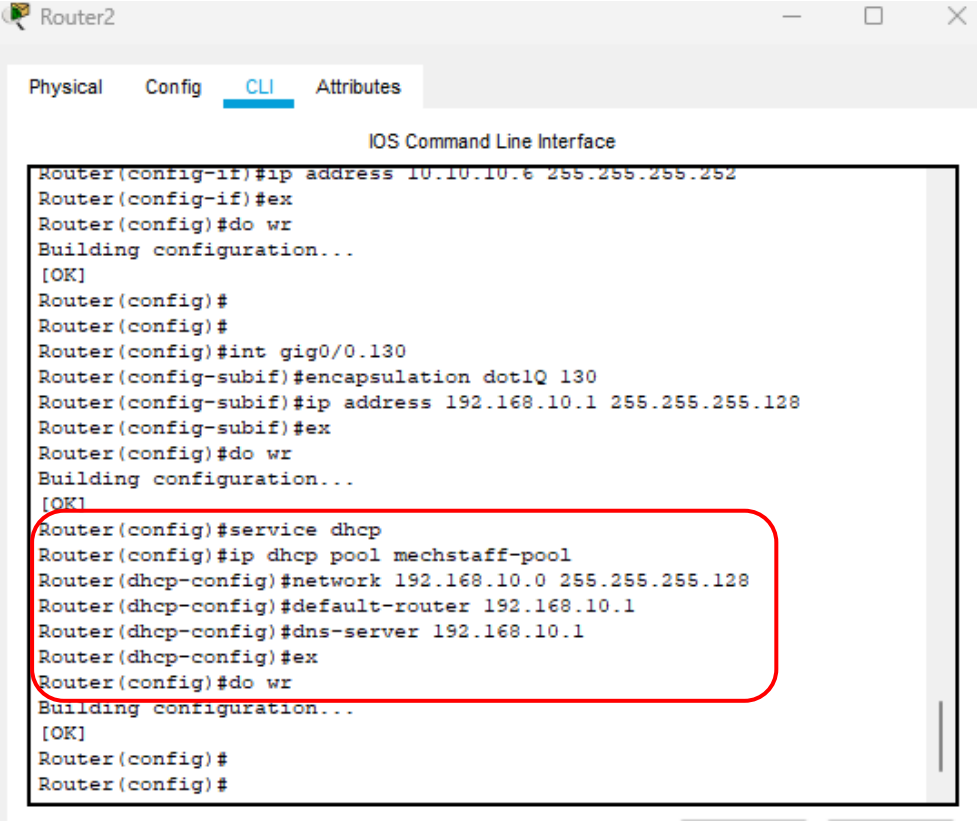
IX. Set the IP for Sub interface of router



The screenshot shows the Router2 CLI interface with the 'CLI' tab selected. The command history shows the configuration of the serial interface se0/1/0. A red box highlights the configuration of the sub-interface gig0/0.130, including encapsulation, IP address, and the 'do wr' command.

```
Router2
Physical Config CLI Attributes
IOS Command Line Interface
Router(config-if)#clock rate 64000
Router(config-if)#ex
Router(config)#do wr
Building configuration...
[OK]
Router(config)#
Router(config)#int se0/1/1
Router(config-if)#ip address 10.10.10.10 255.255.255.252
Router(config-if)#ex
Router(config)#int se0/1/0
Router(config-if)#ip address 10.10.10.6 255.255.255.252
Router(config-if)#ex
Router(config)#do wr
Building configuration...
[OK]
Router(config)#
Router(config)#
Router(config)#int gig0/0.130
Router(config-subif)#encapsulation dot1Q 130
Router(config-subif)#ip address 192.168.10.1 255.255.255.128
Router(config-subif)#ex
Router(config)#do wr
Building configuration...
[OK]
Router(config)#
```

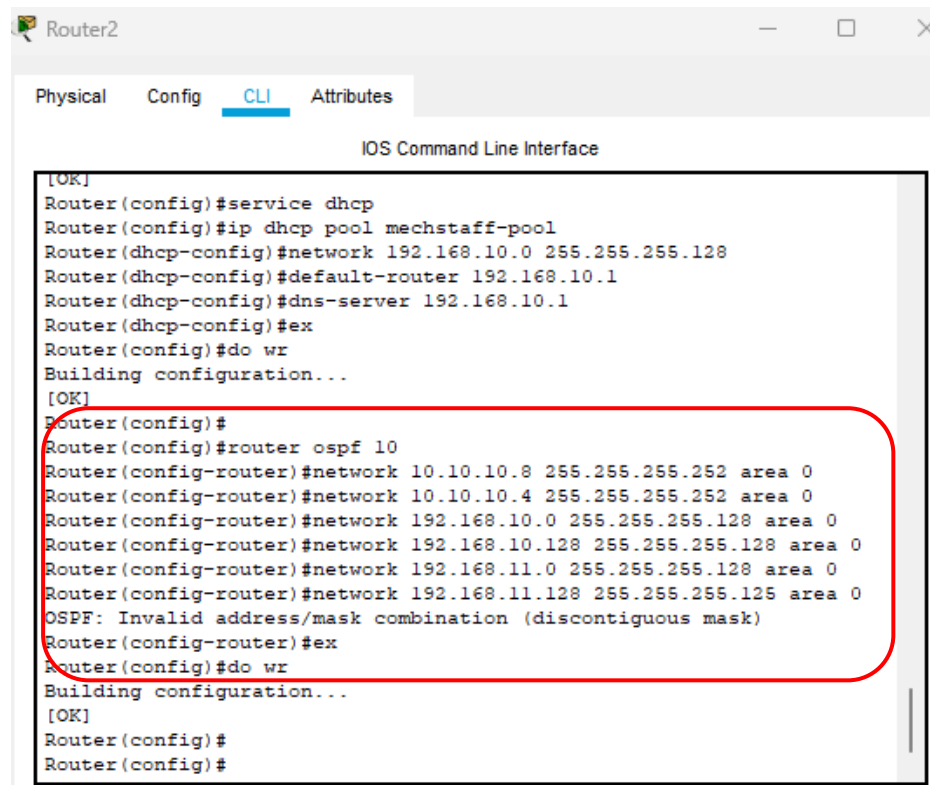
X. Call the DHCP sever to set the dynamic IP



The screenshot shows the Router2 CLI interface with the 'CLI' tab selected. The command history shows the configuration of the sub-interface gig0/0.130 and the DHCP pool. A red box highlights the DHCP configuration, including the 'service dhcp' command, the pool name 'mechstaff-pool', the network, default-router, and dns-server.

```
Router2
Physical Config CLI Attributes
IOS Command Line Interface
Router(config-if)#ip address 10.10.10.6 255.255.255.252
Router(config-if)#ex
Router(config)#do wr
Building configuration...
[OK]
Router(config)#
Router(config)#
Router(config)#int gig0/0.130
Router(config-subif)#encapsulation dot1Q 130
Router(config-subif)#ip address 192.168.10.1 255.255.255.128
Router(config-subif)#ex
Router(config)#do wr
Building configuration...
[OK]
Router(config)#service dhcp
Router(config)#ip dhcp pool mechstaff-pool
Router(dhcp-config)#network 192.168.10.0 255.255.255.128
Router(dhcp-config)#default-router 192.168.10.1
Router(dhcp-config)#dns-server 192.168.10.1
Router(dhcp-config)#ex
Router(config)#do wr
Building configuration...
[OK]
Router(config)#
Router(config)#
```

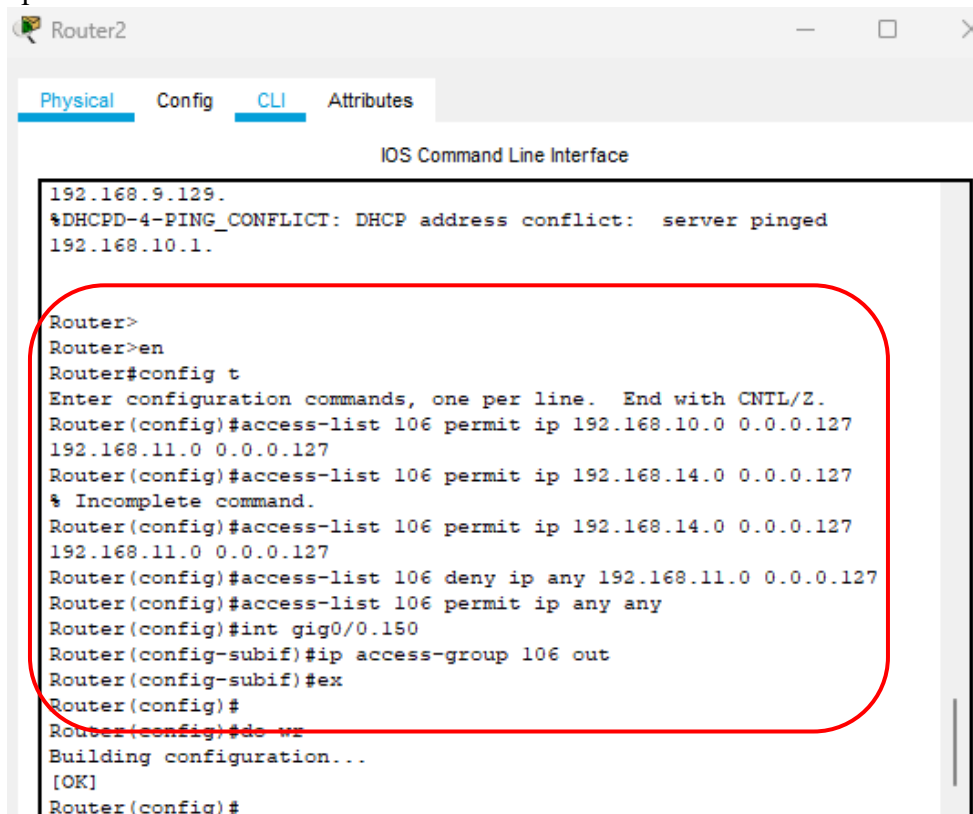
XI. Set the OSPF protocol



```
Router2
Physical Config CLI Attributes
IOS Command Line Interface
[OK]
Router(config)#service dhcp
Router(config)#ip dhcp pool mechstaff-pool
Router(dhcp-config)#network 192.168.10.0 255.255.255.128
Router(dhcp-config)#default-router 192.168.10.1
Router(dhcp-config)#dns-server 192.168.10.1
Router(dhcp-config)#ex
Router(config)#do wr
Building configuration...
[OK]
Router(config)#
Router(config)#router ospf 10
Router(config-router)#network 10.10.10.8 255.255.255.252 area 0
Router(config-router)#network 10.10.10.4 255.255.255.252 area 0
Router(config-router)#network 192.168.10.0 255.255.255.128 area 0
Router(config-router)#network 192.168.10.128 255.255.255.128 area 0
Router(config-router)#network 192.168.11.0 255.255.255.128 area 0
Router(config-router)#network 192.168.11.128 255.255.255.128 area 0
OSPF: Invalid address/mask combination (discontiguous mask)
Router(config-router)#ex
Router(config)#do wr
Building configuration...
[OK]
Router(config)#
Router(config)#
```

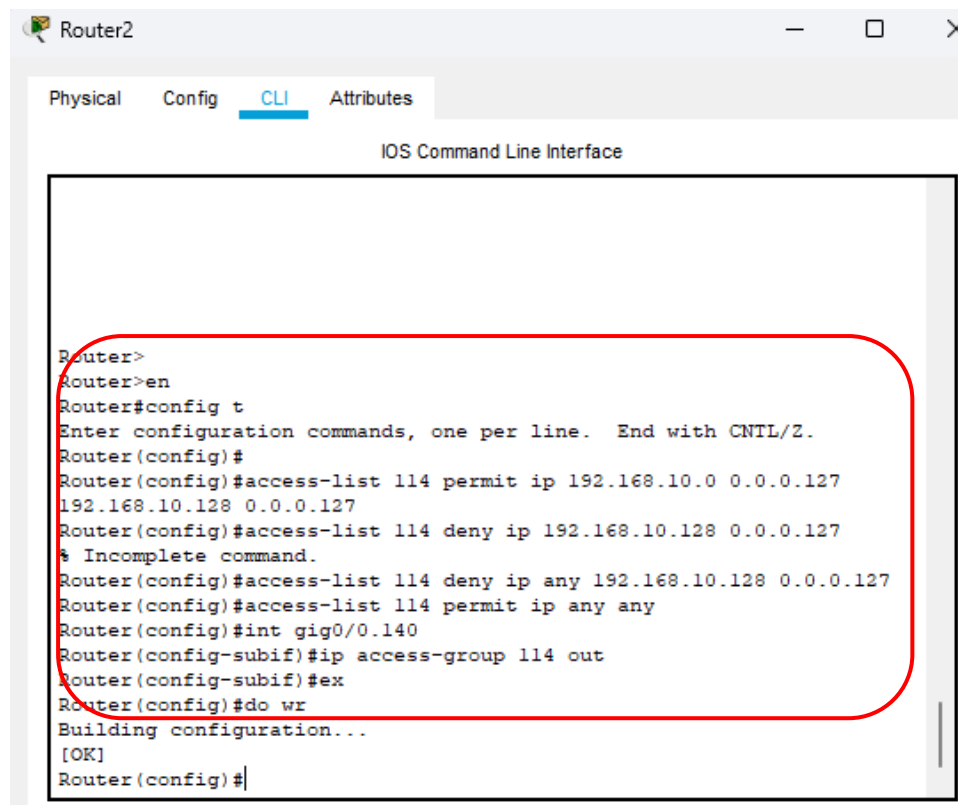
XII. Create an access list

- ✓ Create an access list for printers & shared devices permit only mech-staff & admin-staff to access the printers & shared devices



```
Router2
Physical Config CLI Attributes
IOS Command Line Interface
192.168.9.129.
%DHCPD-4-PING_CONFLICT: DHCP address conflict: server pinged
192.168.10.1.
Router>
Router>en
Router#config t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#access-list 106 permit ip 192.168.10.0 0.0.0.127
192.168.11.0 0.0.0.127
Router(config)#access-list 106 permit ip 192.168.14.0 0.0.0.127
% Incomplete command.
Router(config)#access-list 106 permit ip 192.168.14.0 0.0.0.127
192.168.11.0 0.0.0.127
Router(config)#access-list 106 deny ip any 192.168.11.0 0.0.0.127
Router(config)#access-list 106 permit ip any any
Router(config)#int gig0/0.150
Router(config-subif)#ip access-group 106 out
Router(config-subif)#ex
Router(config)#
Router(config)#do wr
Building configuration...
[OK]
Router(config)#
```

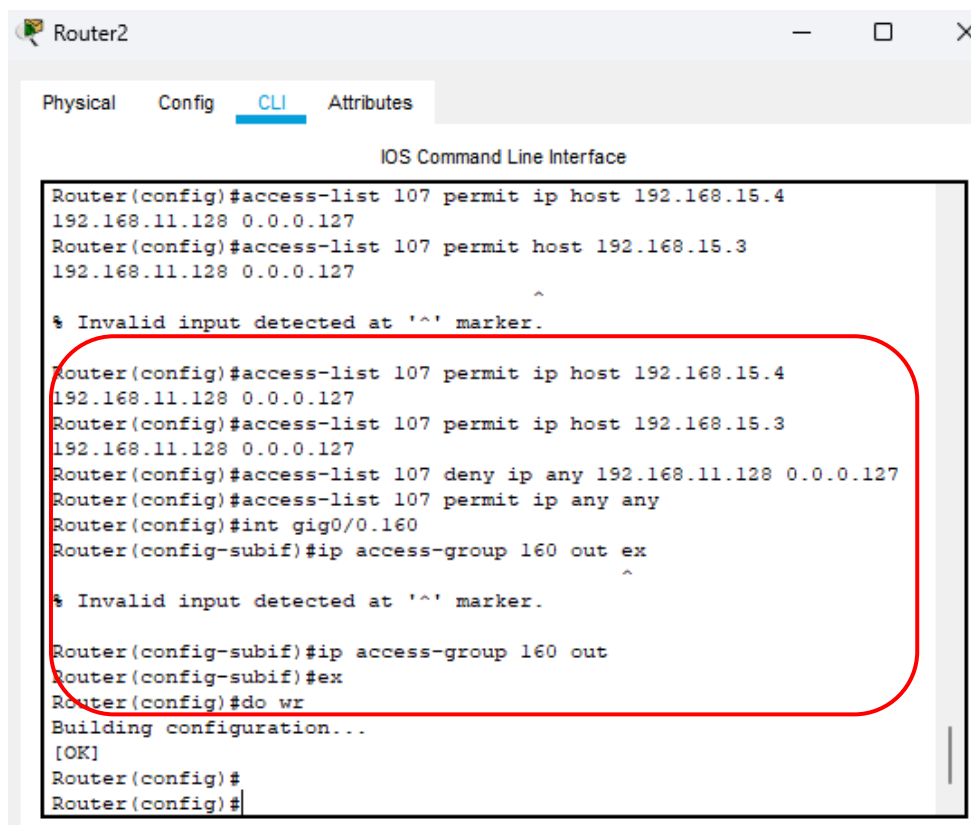

- ✓ Create a access list for student-devices ,permit only mech-staff to access the student-devices



```
Router2
Physical Config CLI Attributes
IOS Command Line Interface

Router>
Router>en
Router#config t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#
Router(config)#access-list 114 permit ip 192.168.10.0 0.0.0.127
192.168.10.128 0.0.0.127
Router(config)#access-list 114 deny ip 192.168.10.128 0.0.0.127
% Incomplete command.
Router(config)#access-list 114 deny ip any 192.168.10.128 0.0.0.127
Router(config)#access-list 114 permit ip any any
Router(config)#int gig0/0.140
Router(config-subif)#ip access-group 114 out
Router(config-subif)#ex
Router(config)#do wr
Building configuration...
[OK]
Router(config)#
```

- ✓ Create a access list for mech- CCTV ,permit only two host of admin's CCTV section to access the mech- CCTV



```
Router2
Physical Config CLI Attributes
IOS Command Line Interface

Router(config)#access-list 107 permit ip host 192.168.15.4
192.168.11.128 0.0.0.127
Router(config)#access-list 107 permit host 192.168.15.3
192.168.11.128 0.0.0.127
^
% Invalid input detected at '^' marker.

Router(config)#access-list 107 permit ip host 192.168.15.4
192.168.11.128 0.0.0.127
Router(config)#access-list 107 permit ip host 192.168.15.3
192.168.11.128 0.0.0.127
Router(config)#access-list 107 deny ip any 192.168.11.128 0.0.0.127
Router(config)#access-list 107 permit ip any any
Router(config)#int gig0/0.160
Router(config-subif)#ip access-group 160 out ex
^
% Invalid input detected at '^' marker.

Router(config-subif)#ip access-group 160 out
Router(config-subif)#ex
Router(config)#do wr
Building configuration...
[OK]
Router(config)#
Router(config)#
```

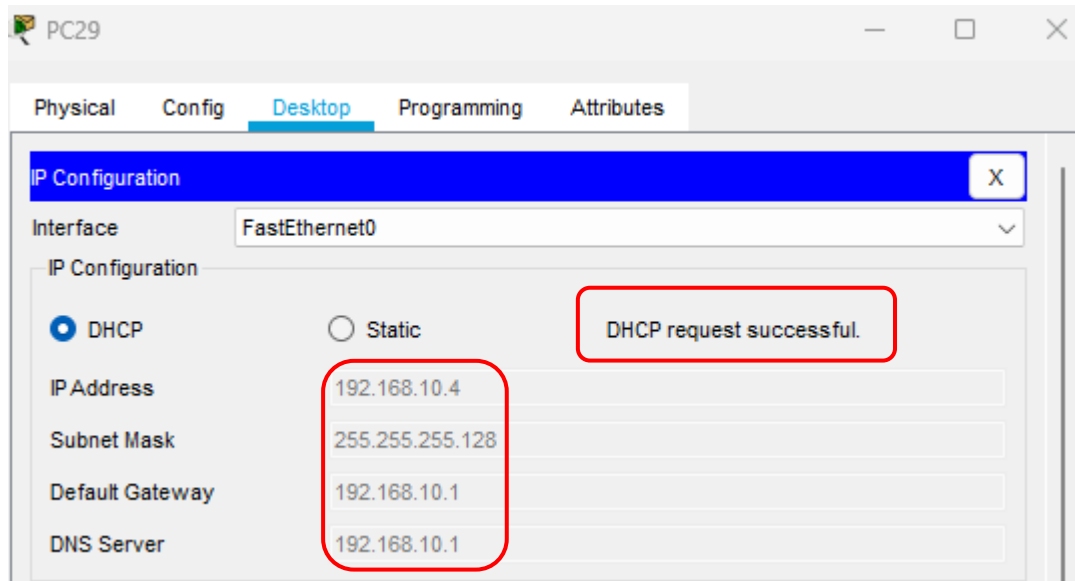
Result of IP DHCP server protocol

- For Department of Mechanical

1. Staff

Subnet-Address 192.168.10.0/25

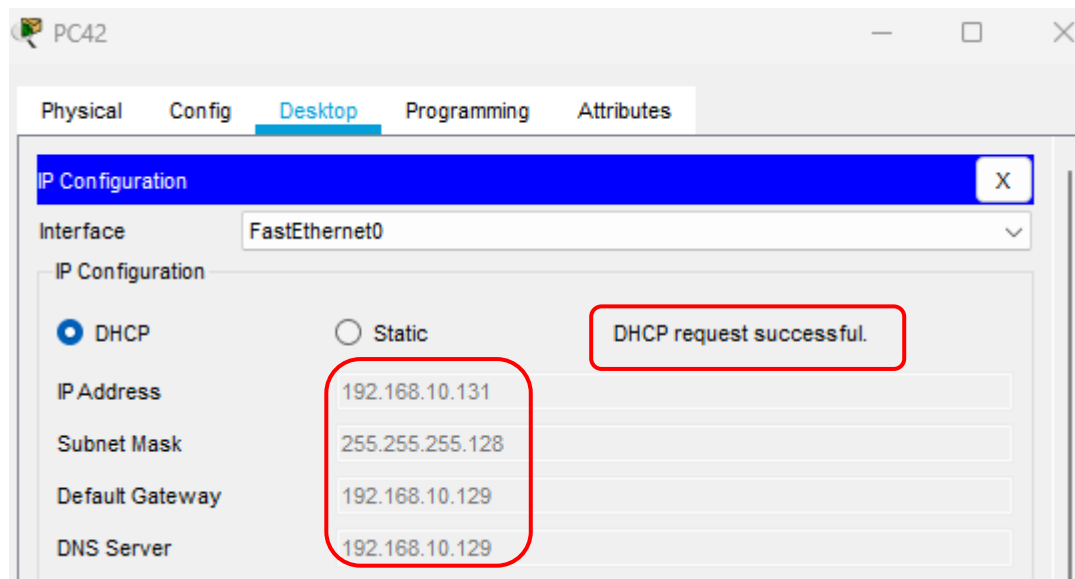
IP range 192.168.10.1-192.168.10.126



2. Student

Subnet-Address 192.168.10.128/25

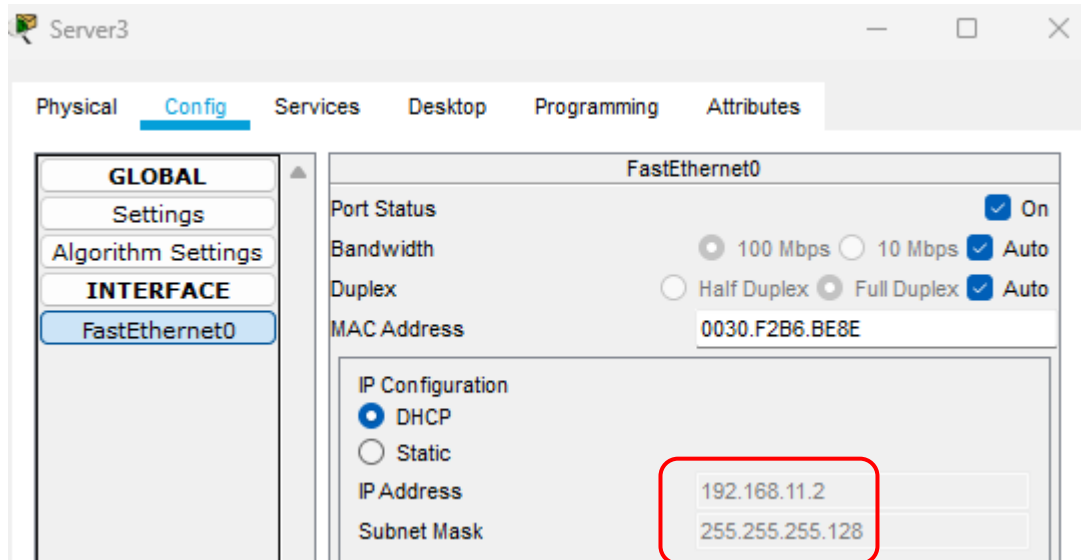
IP range 192.168.10.129-192.168.10.254



3. Printers & Shared Devices

Subnet-Address 192.168.11.0/25

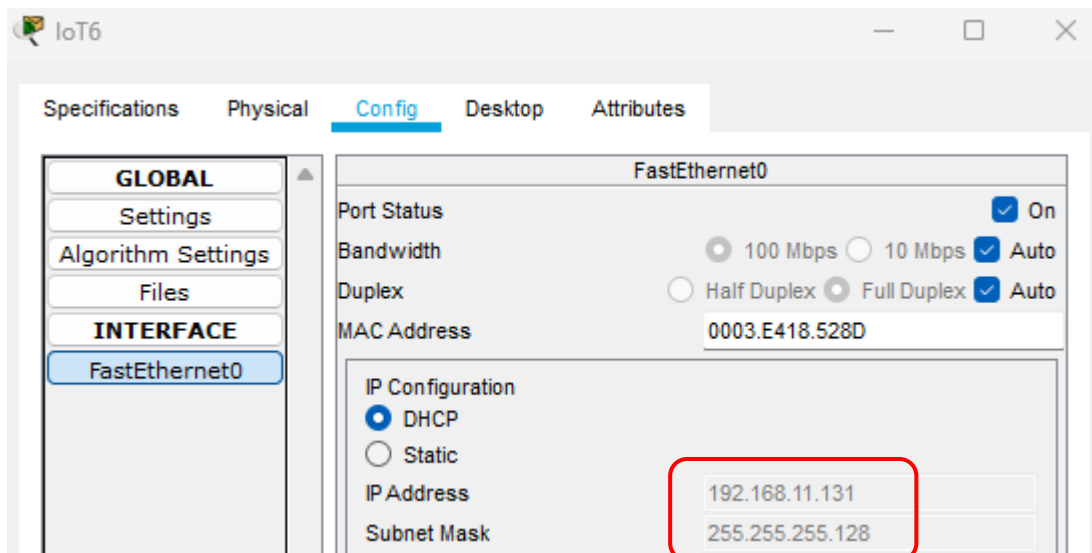
IP range 192.168.11.1-192.168.11.126



4. CCTV

Subnet-Address 192.168.11.128/25

IP range 192.168.11.129-192.168.11.254

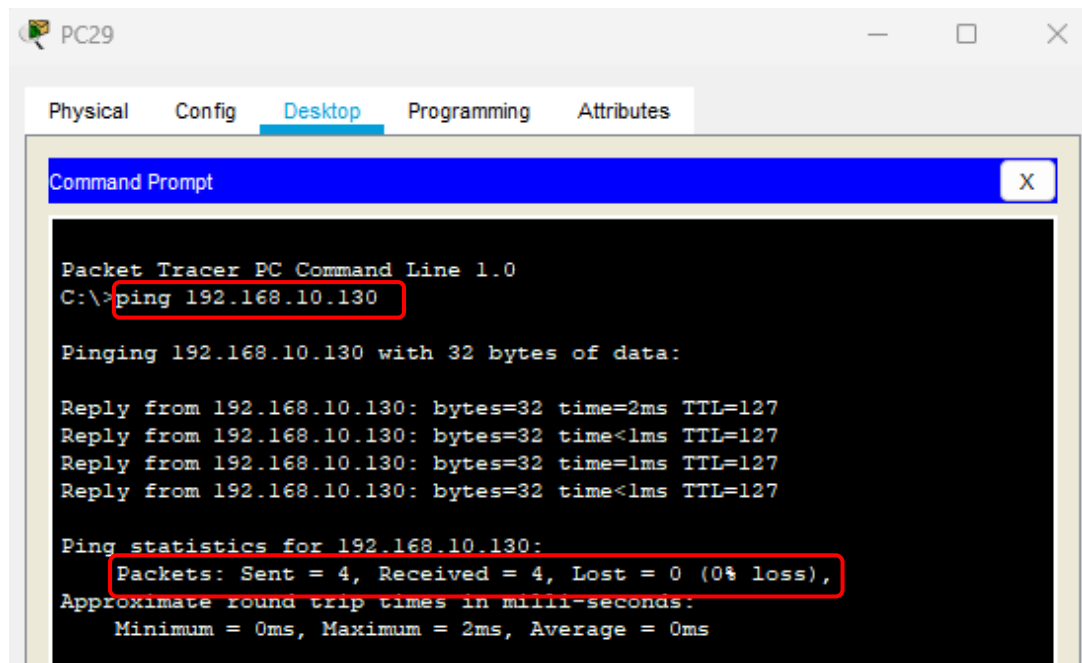


Simulation Results(Device connectivity (ping and traceroute results).)

1. 192.168.10.128/25(Subnet Address of mech-student)

192.168.10.129-192.168.10.254 (Range of mech-student's IP)

- Pinging from mech-staff to mech-student



The screenshot shows a Packet Tracer PC window for PC29. The 'Desktop' tab is active, displaying a Command Prompt. The command 'ping 192.168.10.130' has been entered and executed. The output shows four successful replies from 192.168.10.130 with 32 bytes of data, response times of 2ms or less, and a TTL of 127. The ping statistics summary at the bottom indicates 4 packets sent, 4 received, 0 lost (0% loss), and round trip times of 0ms minimum, 2ms maximum, and 0ms average.

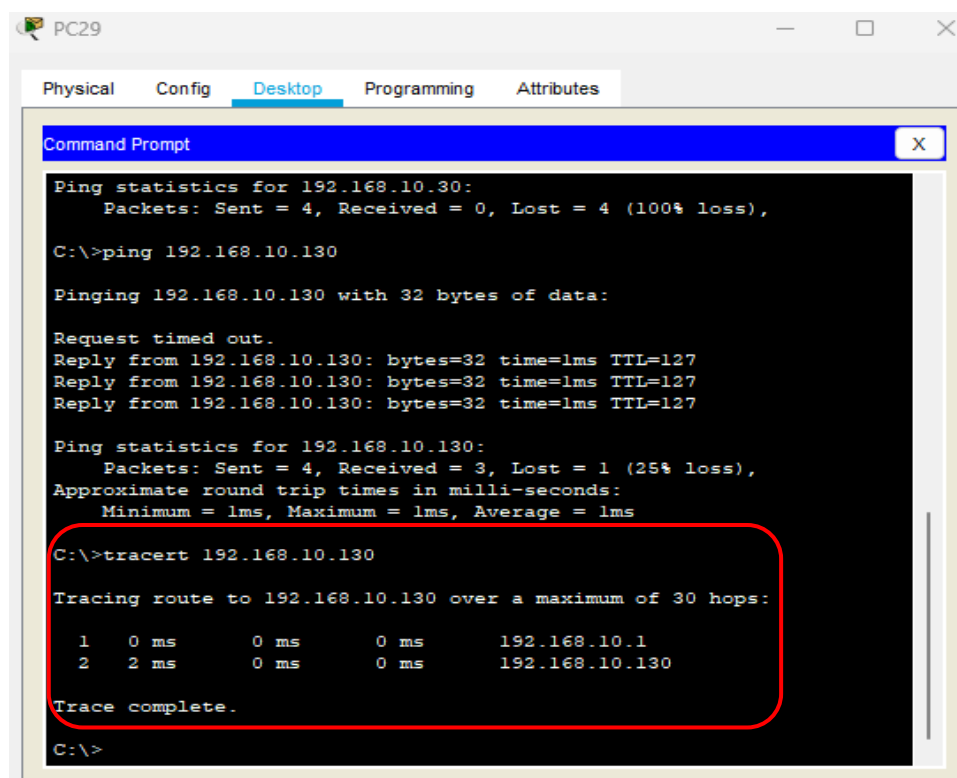
```
Packet Tracer PC Command Line 1.0
C:\>ping 192.168.10.130

Pinging 192.168.10.130 with 32 bytes of data:

Reply from 192.168.10.130: bytes=32 time=2ms TTL=127
Reply from 192.168.10.130: bytes=32 time<1ms TTL=127
Reply from 192.168.10.130: bytes=32 time=1ms TTL=127
Reply from 192.168.10.130: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.10.130:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 2ms, Average = 0ms
```

The mech-staff can access mech-student



The screenshot shows the same Packet Tracer PC window for PC29. The Command Prompt shows a failed ping attempt to 192.168.10.130, with statistics showing 4 packets sent, 0 received, and 4 lost (100% loss). A 'Request timed out' message is displayed. Following this, the command 'tracert 192.168.10.130' is entered and executed. The output shows a successful traceroute with two hops: 192.168.10.1 and 192.168.10.130, both with 0ms response times. The trace is complete.

```
Ping statistics for 192.168.10.130:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>ping 192.168.10.130

Pinging 192.168.10.130 with 32 bytes of data:

Request timed out.
Reply from 192.168.10.130: bytes=32 time=1ms TTL=127
Reply from 192.168.10.130: bytes=32 time=1ms TTL=127
Reply from 192.168.10.130: bytes=32 time=1ms TTL=127

Ping statistics for 192.168.10.130:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 1ms, Average = 1ms

C:\>tracert 192.168.10.130

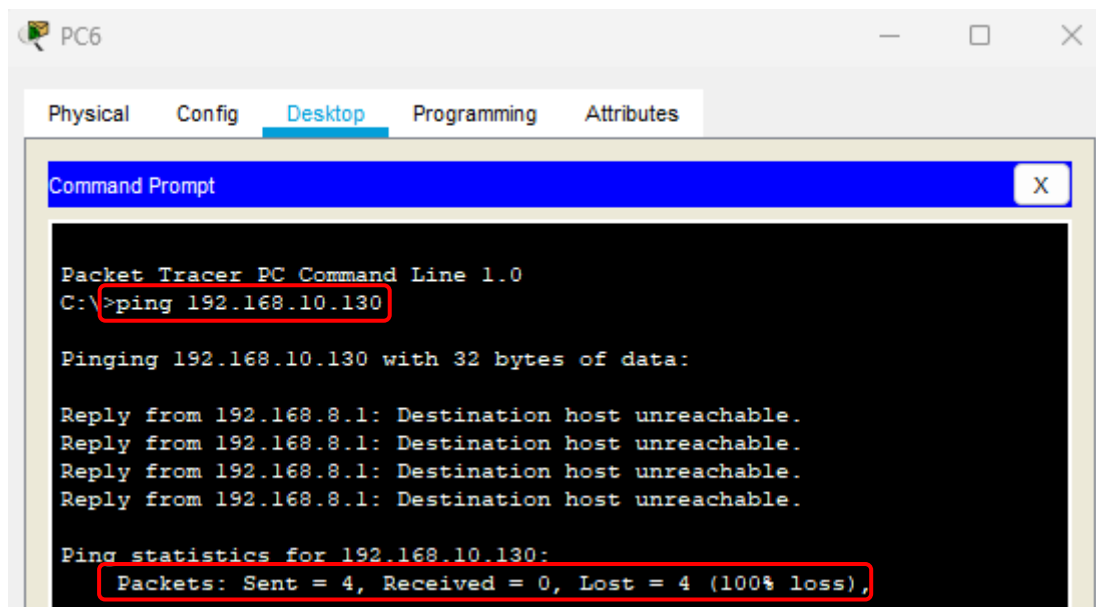
Tracing route to 192.168.10.130 over a maximum of 30 hops:

  1  0 ms    0 ms    0 ms    192.168.10.1
  2  2 ms    0 ms    0 ms    192.168.10.130

Trace complete.

C:\>
```

- Pinging from another department's staff to mech-student



PC6

Physical Config **Desktop** Programming Attributes

Command Prompt

```

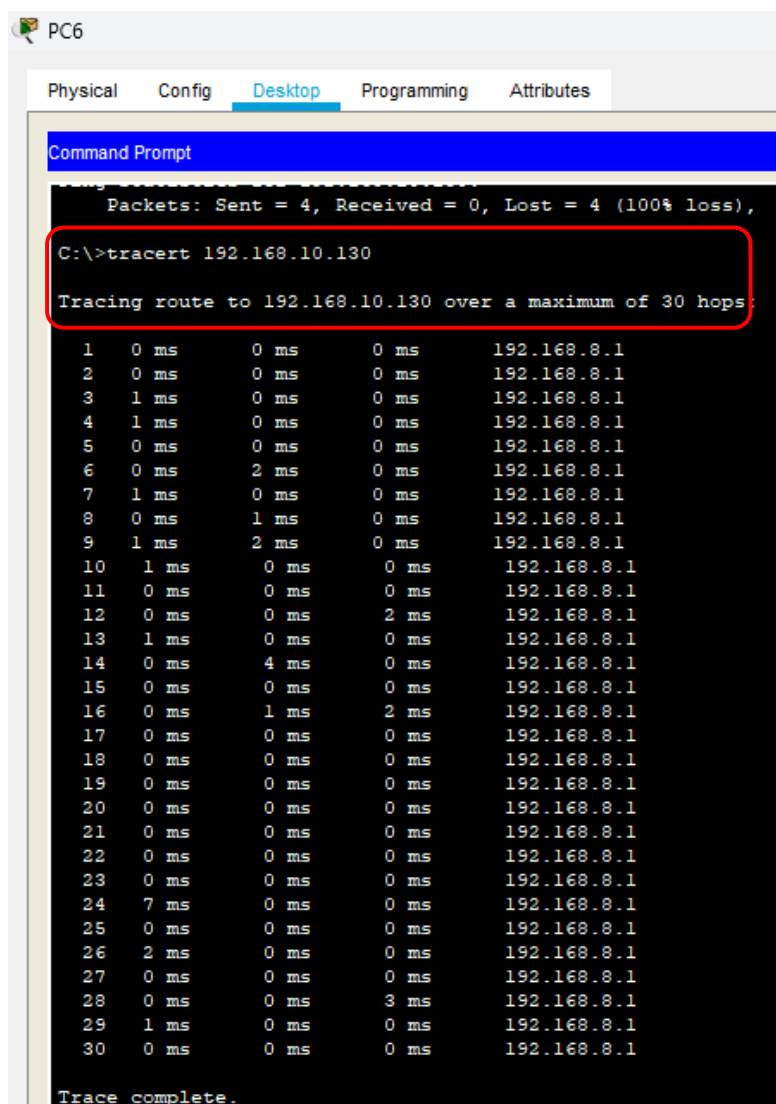
Packet Tracer PC Command Line 1.0
C:\>ping 192.168.10.130

Pinging 192.168.10.130 with 32 bytes of data:

Reply from 192.168.8.1: Destination host unreachable.
Reply from 192.168.8.1: Destination host unreachable.
Reply from 192.168.8.1: Destination host unreachable.
Reply from 192.168.8.1: Destination host unreachable.

Ping statistics for 192.168.10.130:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
  
```

Only mech-staff can access the mech-student not other department's staff



PC6

Physical Config **Desktop** Programming Attributes

Command Prompt

```

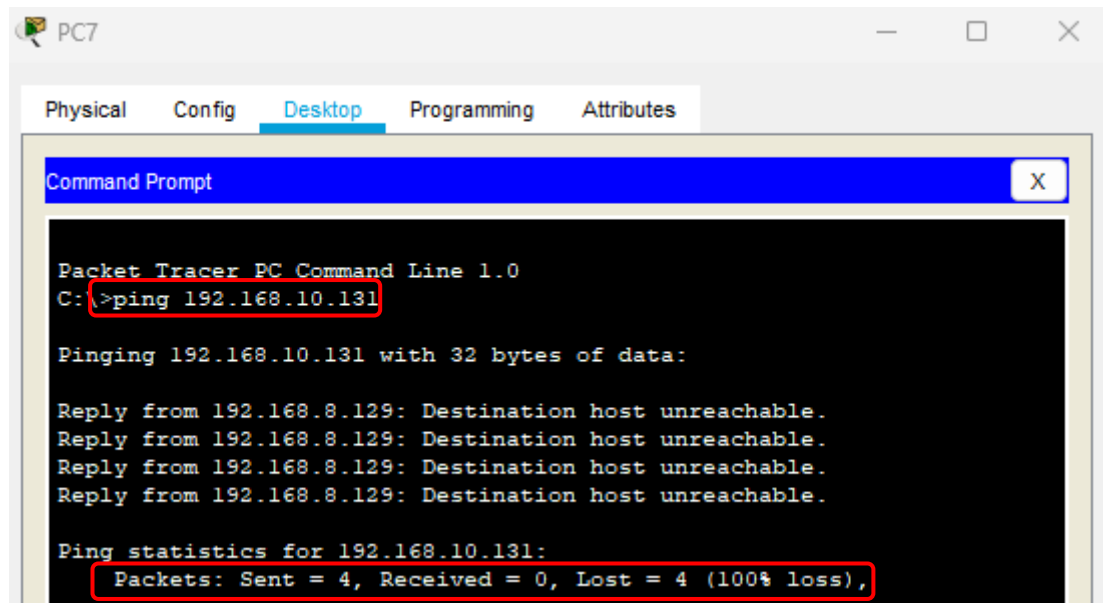
Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
C:\>tracert 192.168.10.130

Tracing route to 192.168.10.130 over a maximum of 30 hops:

  0  0 ms    0 ms    0 ms   192.168.8.1
  1  0 ms    0 ms    0 ms   192.168.8.1
  2  1 ms    0 ms    0 ms   192.168.8.1
  3  1 ms    0 ms    0 ms   192.168.8.1
  4  0 ms    0 ms    0 ms   192.168.8.1
  5  0 ms    2 ms    0 ms   192.168.8.1
  6  1 ms    0 ms    0 ms   192.168.8.1
  7  0 ms    1 ms    0 ms   192.168.8.1
  8  1 ms    2 ms    0 ms   192.168.8.1
  9  1 ms    0 ms    0 ms   192.168.8.1
 10  0 ms    0 ms    0 ms   192.168.8.1
 11  0 ms    0 ms    2 ms   192.168.8.1
 12  1 ms    0 ms    0 ms   192.168.8.1
 13  0 ms    4 ms    0 ms   192.168.8.1
 14  0 ms    0 ms    0 ms   192.168.8.1
 15  0 ms    1 ms    2 ms   192.168.8.1
 16  0 ms    0 ms    0 ms   192.168.8.1
 17  0 ms    0 ms    0 ms   192.168.8.1
 18  0 ms    0 ms    0 ms   192.168.8.1
 19  0 ms    0 ms    0 ms   192.168.8.1
 20  0 ms    0 ms    0 ms   192.168.8.1
 21  0 ms    0 ms    0 ms   192.168.8.1
 22  0 ms    0 ms    0 ms   192.168.8.1
 23  0 ms    0 ms    0 ms   192.168.8.1
 24  7 ms    0 ms    0 ms   192.168.8.1
 25  2 ms    0 ms    0 ms   192.168.8.1
 26  0 ms    0 ms    0 ms   192.168.8.1
 27  0 ms    3 ms    0 ms   192.168.8.1
 28  1 ms    0 ms    0 ms   192.168.8.1
 29  0 ms    0 ms    0 ms   192.168.8.1
 30  0 ms    0 ms    0 ms   192.168.8.1

Trace complete.
  
```

- Pinging from another departments student to mech-student



The screenshot shows the Packet Tracer interface for PC7. The 'Desktop' tab is selected, displaying a 'Command Prompt' window. The command prompt shows the execution of the command 'ping 192.168.10.131'. The output indicates that the destination host is unreachable, with a 100% loss of packets.

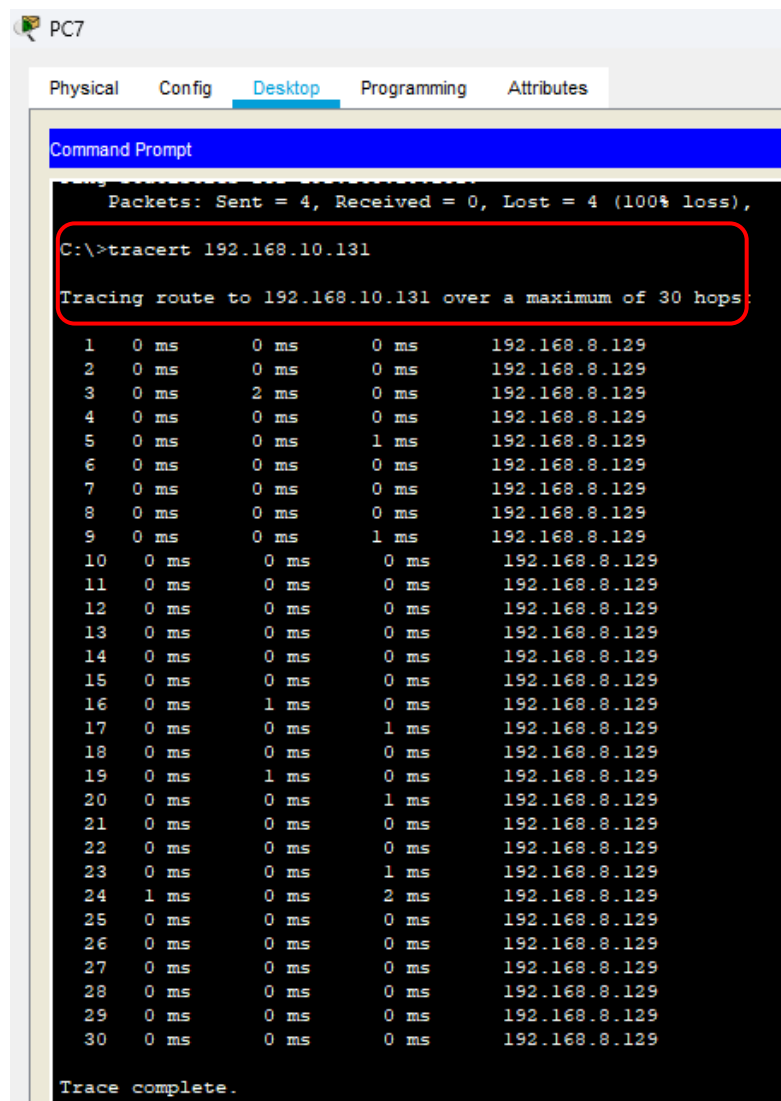
```
Packet Tracer PC Command Line 1.0
C:\>ping 192.168.10.131

Pinging 192.168.10.131 with 32 bytes of data:

Reply from 192.168.8.129: Destination host unreachable.
Reply from 192.168.8.129: Destination host unreachable.
Reply from 192.168.8.129: Destination host unreachable.
Reply from 192.168.8.129: Destination host unreachable.

Ping statistics for 192.168.10.131:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

Can't access , only mech-staff can access the mech-student



The screenshot shows the Packet Tracer interface for PC7. The 'Desktop' tab is selected, displaying a 'Command Prompt' window. The command prompt shows the execution of the command 'tracert 192.168.10.131'. The output shows the route taken by the packets, with all hops reaching the destination.

```
Packet Tracer PC Command Line 1.0
C:\>tracert 192.168.10.131

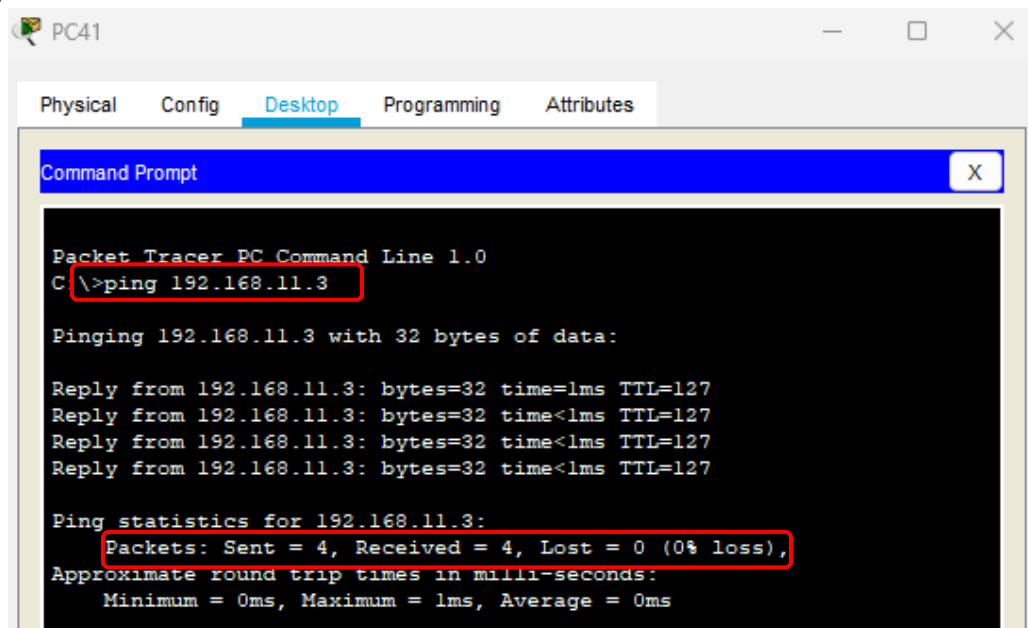
Tracing route to 192.168.10.131 over a maximum of 30 hops:

  0  0 ms    0 ms    0 ms    192.168.8.129
  1  0 ms    0 ms    0 ms    192.168.8.129
  2  0 ms    2 ms    0 ms    192.168.8.129
  3  0 ms    0 ms    0 ms    192.168.8.129
  4  0 ms    0 ms    1 ms    192.168.8.129
  5  0 ms    0 ms    0 ms    192.168.8.129
  6  0 ms    0 ms    0 ms    192.168.8.129
  7  0 ms    0 ms    0 ms    192.168.8.129
  8  0 ms    0 ms    0 ms    192.168.8.129
  9  0 ms    0 ms    1 ms    192.168.8.129
 10  0 ms    0 ms    0 ms    192.168.8.129
 11  0 ms    0 ms    0 ms    192.168.8.129
 12  0 ms    0 ms    0 ms    192.168.8.129
 13  0 ms    0 ms    0 ms    192.168.8.129
 14  0 ms    0 ms    0 ms    192.168.8.129
 15  0 ms    0 ms    0 ms    192.168.8.129
 16  0 ms    1 ms    0 ms    192.168.8.129
 17  0 ms    0 ms    1 ms    192.168.8.129
 18  0 ms    0 ms    0 ms    192.168.8.129
 19  0 ms    1 ms    0 ms    192.168.8.129
 20  0 ms    0 ms    1 ms    192.168.8.129
 21  0 ms    0 ms    0 ms    192.168.8.129
 22  0 ms    0 ms    0 ms    192.168.8.129
 23  0 ms    0 ms    1 ms    192.168.8.129
 24  1 ms    0 ms    2 ms    192.168.8.129
 25  0 ms    0 ms    0 ms    192.168.8.129
 26  0 ms    0 ms    0 ms    192.168.8.129
 27  0 ms    0 ms    0 ms    192.168.8.129
 28  0 ms    0 ms    0 ms    192.168.8.129
 29  0 ms    0 ms    0 ms    192.168.8.129
 30  0 ms    0 ms    0 ms    192.168.8.129

Trace complete.
```

2. 192.168.11.0/25(Subnet Address of mech-Printers & Shared Devices) 192.168.11.1-192.168.11.126 (Range of mech-Printers & Shared Devices IP)

- Pinging from mech-staff to mech-Printers & Shared Devices



The screenshot shows a Packet Tracer PC window for PC41 with the 'Desktop' tab selected. A Command Prompt window is open, displaying the output of a ping command to 192.168.11.3. The output shows four successful replies with 32 bytes of data, a time of 1ms, and a TTL of 127. The ping statistics show 4 packets sent, 4 received, and 0% loss.

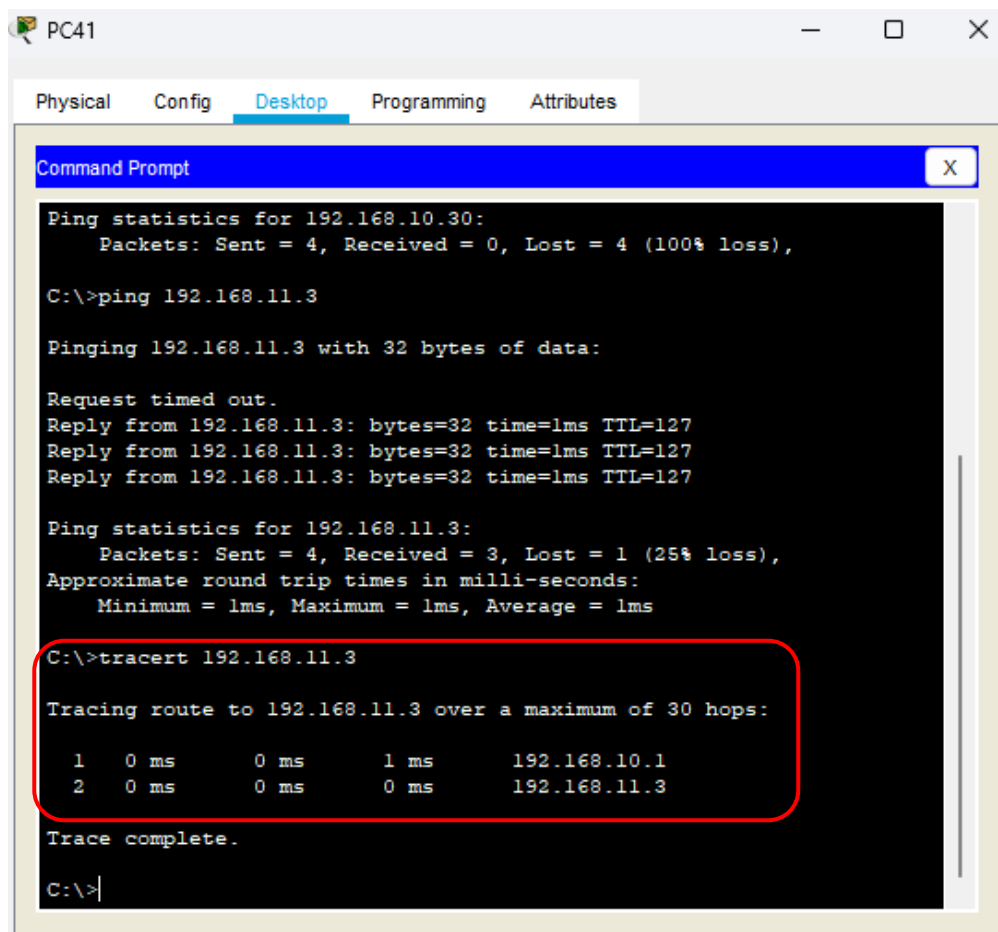
```
Packet Tracer PC Command Line 1.0
C:\>ping 192.168.11.3

Pinging 192.168.11.3 with 32 bytes of data:

Reply from 192.168.11.3: bytes=32 time=1ms TTL=127
Reply from 192.168.11.3: bytes=32 time<1ms TTL=127
Reply from 192.168.11.3: bytes=32 time<1ms TTL=127
Reply from 192.168.11.3: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.11.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms
```

Mech-staff can access the mech-Printers & Shared Devices



The screenshot shows a Packet Tracer PC window for PC41 with the 'Desktop' tab selected. A Command Prompt window is open, displaying the output of a ping command to 192.168.10.30, which failed with 100% loss. Then, a ping command to 192.168.11.3 was entered, but it timed out. Finally, a traceroute command to 192.168.11.3 was entered, showing a successful path from 192.168.10.1 to 192.168.11.3.

```
Packet Tracer PC Command Line 1.0
C:\>ping 192.168.10.30
Ping statistics for 192.168.10.30:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>ping 192.168.11.3
Pinging 192.168.11.3 with 32 bytes of data:

Request timed out.
Reply from 192.168.11.3: bytes=32 time=1ms TTL=127
Reply from 192.168.11.3: bytes=32 time=1ms TTL=127
Reply from 192.168.11.3: bytes=32 time=1ms TTL=127

Ping statistics for 192.168.11.3:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 1ms, Average = 1ms

C:\>tracert 192.168.11.3

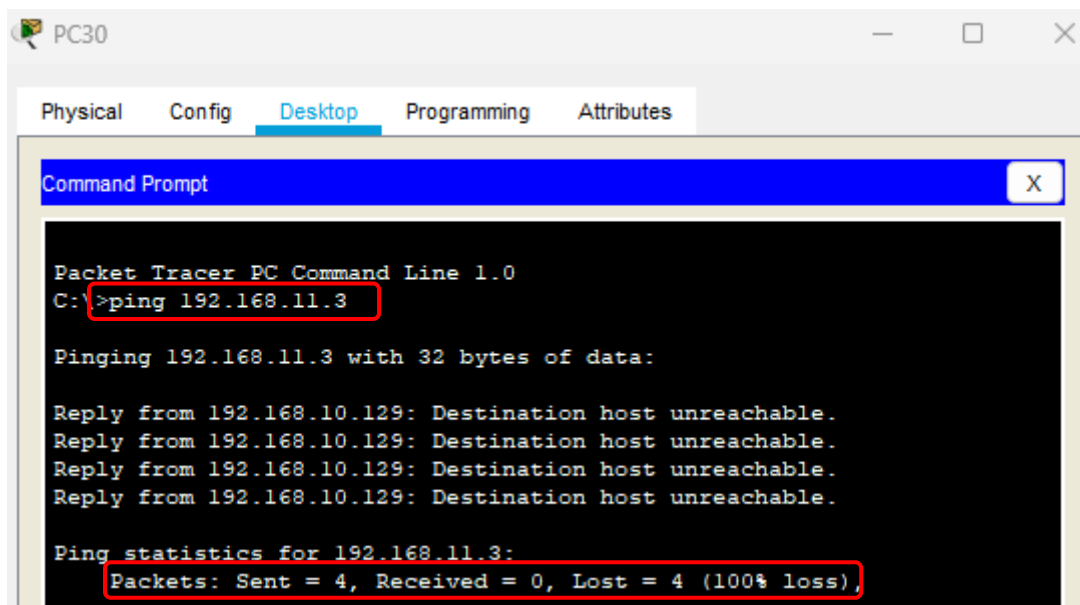
Tracing route to 192.168.11.3 over a maximum of 30 hops:

  1  0 ms    0 ms    1 ms    192.168.10.1
  2  0 ms    0 ms    0 ms    192.168.11.3

Trace complete.

C:\>
```

- Pinging from mech-student to mech mech-Printers & Shared Devices



The screenshot shows the Packet Tracer interface for PC30. The 'Desktop' tab is selected, and the 'Command Prompt' window is open. The command prompt shows the command `C:\>ping 192.168.11.3` and the resulting output. The output indicates that the destination host is unreachable and that all four packets sent were lost.

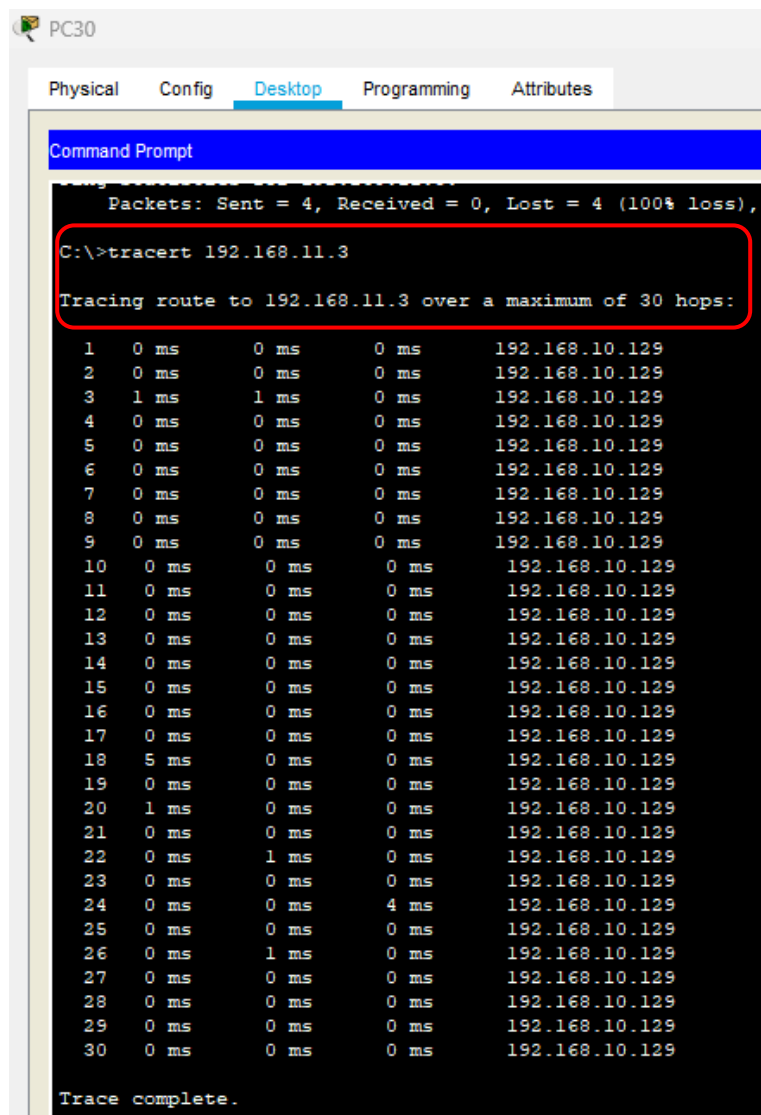
```
Packet Tracer PC Command Line 1.0
C:\>ping 192.168.11.3

Pinging 192.168.11.3 with 32 bytes of data:

Reply from 192.168.10.129: Destination host unreachable.
Reply from 192.168.10.129: Destination host unreachable.
Reply from 192.168.10.129: Destination host unreachable.
Reply from 192.168.10.129: Destination host unreachable.

Ping statistics for 192.168.11.3:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

Only mech-staff can access mech-Printers & Shared Devices



The screenshot shows the Packet Tracer interface for PC30. The 'Desktop' tab is selected, and the 'Command Prompt' window is open. The command prompt shows the command `C:\>tracert 192.168.11.3` and the resulting output. The output shows a successful traceroute to 192.168.11.3 over a maximum of 30 hops.

```
Packet Tracer PC Command Line 1.0
C:\>tracert 192.168.11.3

Tracing route to 192.168.11.3 over a maximum of 30 hops:

  0  0 ms    0 ms    0 ms    192.168.10.129
  1  0 ms    0 ms    0 ms    192.168.10.129
  2  0 ms    0 ms    0 ms    192.168.10.129
  3  1 ms    1 ms    0 ms    192.168.10.129
  4  0 ms    0 ms    0 ms    192.168.10.129
  5  0 ms    0 ms    0 ms    192.168.10.129
  6  0 ms    0 ms    0 ms    192.168.10.129
  7  0 ms    0 ms    0 ms    192.168.10.129
  8  0 ms    0 ms    0 ms    192.168.10.129
  9  0 ms    0 ms    0 ms    192.168.10.129
 10  0 ms    0 ms    0 ms    192.168.10.129
 11  0 ms    0 ms    0 ms    192.168.10.129
 12  0 ms    0 ms    0 ms    192.168.10.129
 13  0 ms    0 ms    0 ms    192.168.10.129
 14  0 ms    0 ms    0 ms    192.168.10.129
 15  0 ms    0 ms    0 ms    192.168.10.129
 16  0 ms    0 ms    0 ms    192.168.10.129
 17  0 ms    0 ms    0 ms    192.168.10.129
 18  5 ms    0 ms    0 ms    192.168.10.129
 19  0 ms    0 ms    0 ms    192.168.10.129
 20  1 ms    0 ms    0 ms    192.168.10.129
 21  0 ms    0 ms    0 ms    192.168.10.129
 22  0 ms    1 ms    0 ms    192.168.10.129
 23  0 ms    0 ms    0 ms    192.168.10.129
 24  0 ms    0 ms    4 ms    192.168.10.129
 25  0 ms    0 ms    0 ms    192.168.10.129
 26  0 ms    1 ms    0 ms    192.168.10.129
 27  0 ms    0 ms    0 ms    192.168.10.129
 28  0 ms    0 ms    0 ms    192.168.10.129
 29  0 ms    0 ms    0 ms    192.168.10.129
 30  0 ms    0 ms    0 ms    192.168.10.129

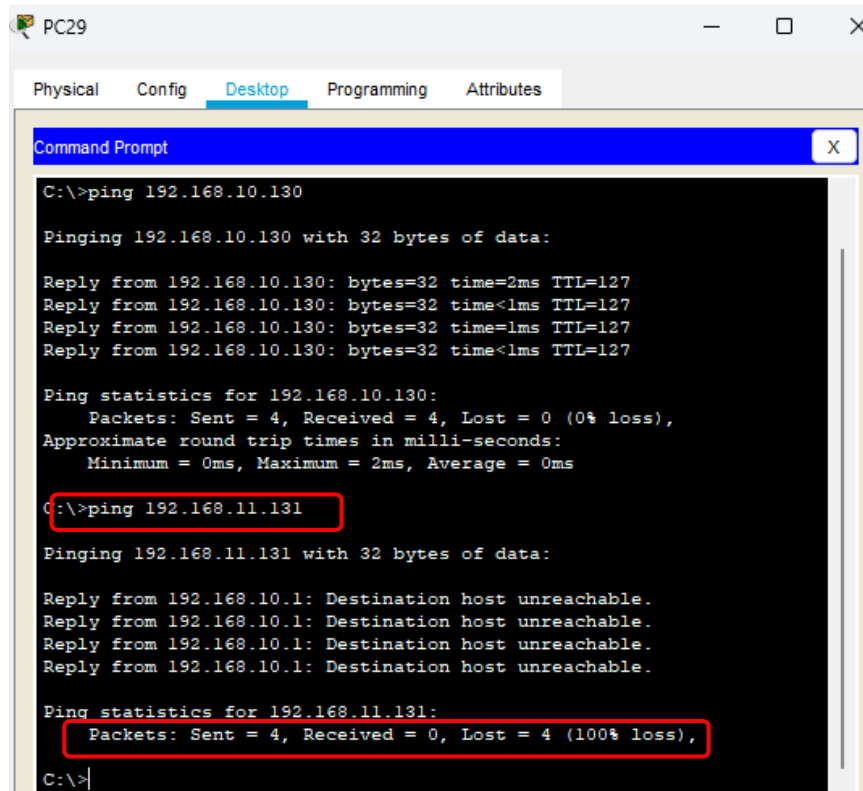
Trace complete.
```


3. 192.168.11.128/25 (Subnet Address of mech-CCTV)

192.168.11.129-192.168.11.254 (Range of mech-CCTV's IP)

- Pinging from mech-staff & admin-CCTV hosts students to mech-CCTV

I. From the mech-staff



The screenshot shows a Windows Command Prompt window titled 'PC29'. The window has tabs for 'Physical', 'Config', 'Desktop' (selected), 'Programming', and 'Attributes'. The Command Prompt shows the following output:

```
C:\>ping 192.168.10.130

Pinging 192.168.10.130 with 32 bytes of data:

Reply from 192.168.10.130: bytes=32 time=2ms TTL=127
Reply from 192.168.10.130: bytes=32 time<1ms TTL=127
Reply from 192.168.10.130: bytes=32 time=1ms TTL=127
Reply from 192.168.10.130: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.10.130:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 2ms, Average = 0ms

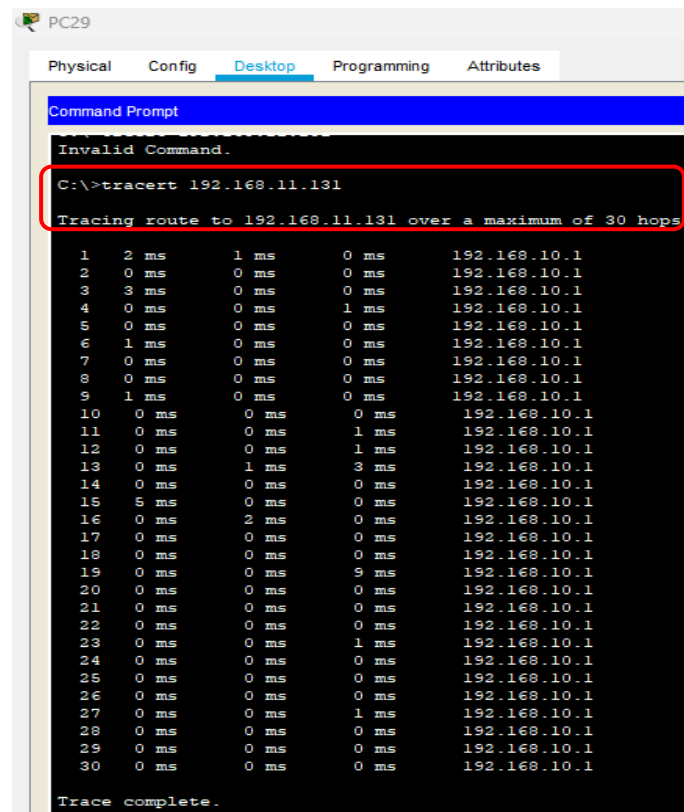
C:\>ping 192.168.11.131

Pinging 192.168.11.131 with 32 bytes of data:

Reply from 192.168.10.1: Destination host unreachable.
Reply from 192.168.10.1: Destination host unreachable.
Reply from 192.168.10.1: Destination host unreachable.
Reply from 192.168.10.1: Destination host unreachable.

Ping statistics for 192.168.11.131:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

Mech-staff can't access mech CCTV



The screenshot shows a Windows Command Prompt window titled 'PC29'. The window has tabs for 'Physical', 'Config', 'Desktop' (selected), 'Programming', and 'Attributes'. The Command Prompt shows the following output:

```
Invalid Command.

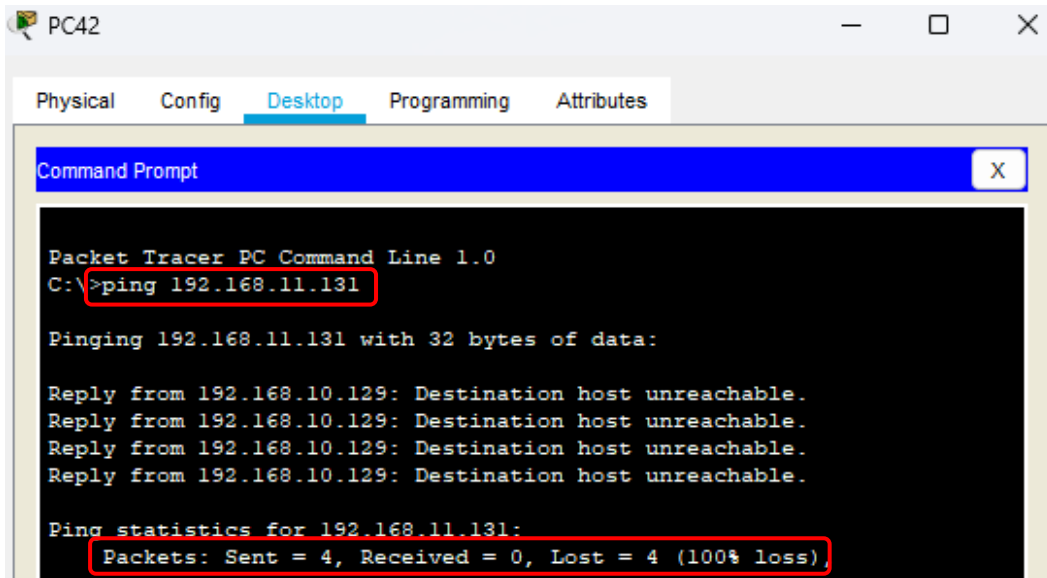
C:\>tracert 192.168.11.131

Tracing route to 192.168.11.131 over a maximum of 30 hops:

  1  2 ms    1 ms     0 ms    192.168.10.1
  2  0 ms    0 ms     0 ms    192.168.10.1
  3  3 ms    0 ms     0 ms    192.168.10.1
  4  0 ms    0 ms     1 ms    192.168.10.1
  5  0 ms    0 ms     0 ms    192.168.10.1
  6  1 ms    0 ms     0 ms    192.168.10.1
  7  0 ms    0 ms     0 ms    192.168.10.1
  8  0 ms    0 ms     0 ms    192.168.10.1
  9  1 ms    0 ms     0 ms    192.168.10.1
 10  0 ms    0 ms     0 ms    192.168.10.1
 11  0 ms    0 ms     1 ms    192.168.10.1
 12  0 ms    0 ms     1 ms    192.168.10.1
 13  0 ms    1 ms     3 ms    192.168.10.1
 14  0 ms    0 ms     0 ms    192.168.10.1
 15  5 ms    0 ms     0 ms    192.168.10.1
 16  0 ms    2 ms     0 ms    192.168.10.1
 17  0 ms    0 ms     0 ms    192.168.10.1
 18  0 ms    0 ms     0 ms    192.168.10.1
 19  0 ms    0 ms     9 ms    192.168.10.1
 20  0 ms    0 ms     0 ms    192.168.10.1
 21  0 ms    0 ms     0 ms    192.168.10.1
 22  0 ms    0 ms     0 ms    192.168.10.1
 23  0 ms    0 ms     1 ms    192.168.10.1
 24  0 ms    0 ms     0 ms    192.168.10.1
 25  0 ms    0 ms     0 ms    192.168.10.1
 26  0 ms    0 ms     0 ms    192.168.10.1
 27  0 ms    0 ms     1 ms    192.168.10.1
 28  0 ms    0 ms     0 ms    192.168.10.1
 29  0 ms    0 ms     0 ms    192.168.10.1
 30  0 ms    0 ms     0 ms    192.168.10.1

Trace complete.
```

II. From the mech-student



PC42

Physical Config **Desktop** Programming Attributes

Command Prompt

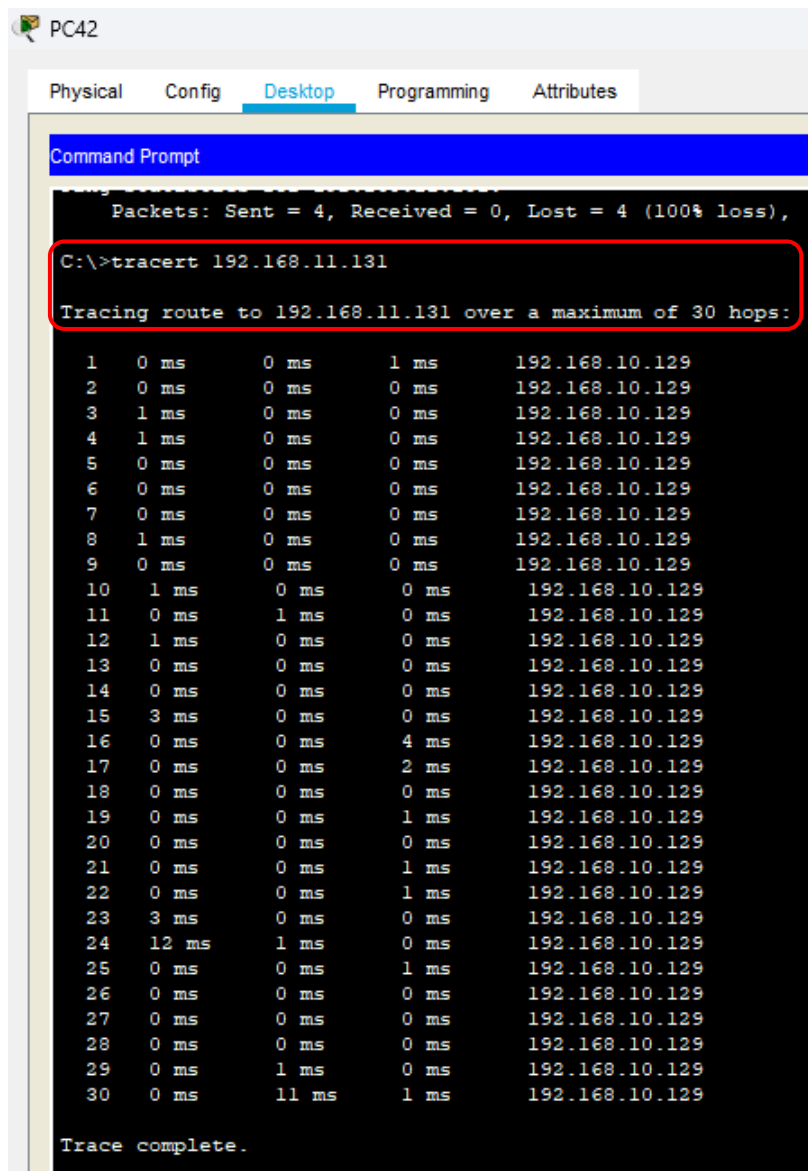
```
Packet Tracer PC Command Line 1.0
C:\>ping 192.168.11.131

Pinging 192.168.11.131 with 32 bytes of data:

Reply from 192.168.10.129: Destination host unreachable.
Reply from 192.168.10.129: Destination host unreachable.
Reply from 192.168.10.129: Destination host unreachable.
Reply from 192.168.10.129: Destination host unreachable.

Ping statistics for 192.168.11.131:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

Mech-students can't access mech-CCTV



PC42

Physical Config **Desktop** Programming Attributes

Command Prompt

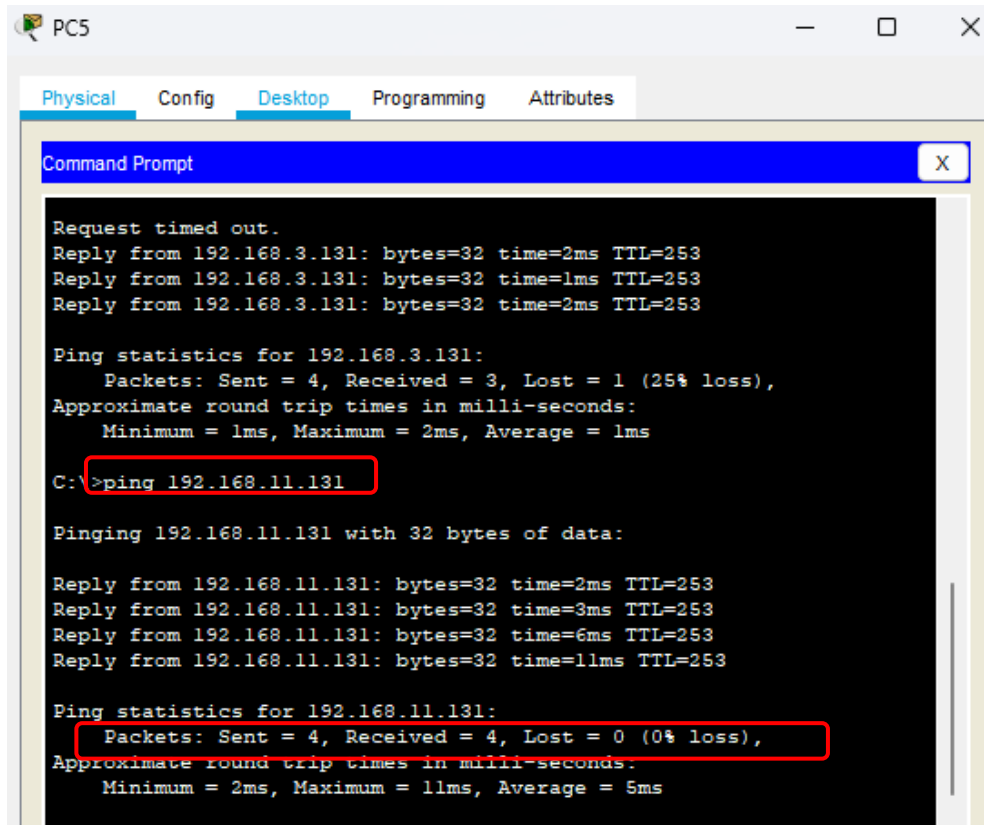
```
Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
C:\>tracert 192.168.11.131

Tracing route to 192.168.11.131 over a maximum of 30 hops:

  1  0 ms    0 ms    1 ms    192.168.10.129
  2  0 ms    0 ms    0 ms    192.168.10.129
  3  1 ms    0 ms    0 ms    192.168.10.129
  4  1 ms    0 ms    0 ms    192.168.10.129
  5  0 ms    0 ms    0 ms    192.168.10.129
  6  0 ms    0 ms    0 ms    192.168.10.129
  7  0 ms    0 ms    0 ms    192.168.10.129
  8  1 ms    0 ms    0 ms    192.168.10.129
  9  0 ms    0 ms    0 ms    192.168.10.129
 10  1 ms    0 ms    0 ms    192.168.10.129
 11  0 ms    1 ms    0 ms    192.168.10.129
 12  1 ms    0 ms    0 ms    192.168.10.129
 13  0 ms    0 ms    0 ms    192.168.10.129
 14  0 ms    0 ms    0 ms    192.168.10.129
 15  3 ms    0 ms    0 ms    192.168.10.129
 16  0 ms    0 ms    4 ms    192.168.10.129
 17  0 ms    0 ms    2 ms    192.168.10.129
 18  0 ms    0 ms    0 ms    192.168.10.129
 19  0 ms    0 ms    1 ms    192.168.10.129
 20  0 ms    0 ms    0 ms    192.168.10.129
 21  0 ms    0 ms    1 ms    192.168.10.129
 22  0 ms    0 ms    1 ms    192.168.10.129
 23  3 ms    0 ms    0 ms    192.168.10.129
 24 12 ms    1 ms    0 ms    192.168.10.129
 25  0 ms    0 ms    1 ms    192.168.10.129
 26  0 ms    0 ms    0 ms    192.168.10.129
 27  0 ms    0 ms    0 ms    192.168.10.129
 28  0 ms    0 ms    0 ms    192.168.10.129
 29  0 ms    1 ms    0 ms    192.168.10.129
 30  0 ms    11 ms   1 ms    192.168.10.129

Trace complete.
```

III. From the hosts of admin CCTV section



The screenshot shows a Windows PC5 desktop with a Command Prompt window open. The window title is "Command Prompt". The desktop has tabs for "Physical", "Config", "Desktop", "Programming", and "Attributes". The Command Prompt shows the following output:

```
Request timed out.
Reply from 192.168.3.131: bytes=32 time=2ms TTL=253
Reply from 192.168.3.131: bytes=32 time=1ms TTL=253
Reply from 192.168.3.131: bytes=32 time=2ms TTL=253

Ping statistics for 192.168.3.131:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 2ms, Average = 1ms

C:\>ping 192.168.11.131

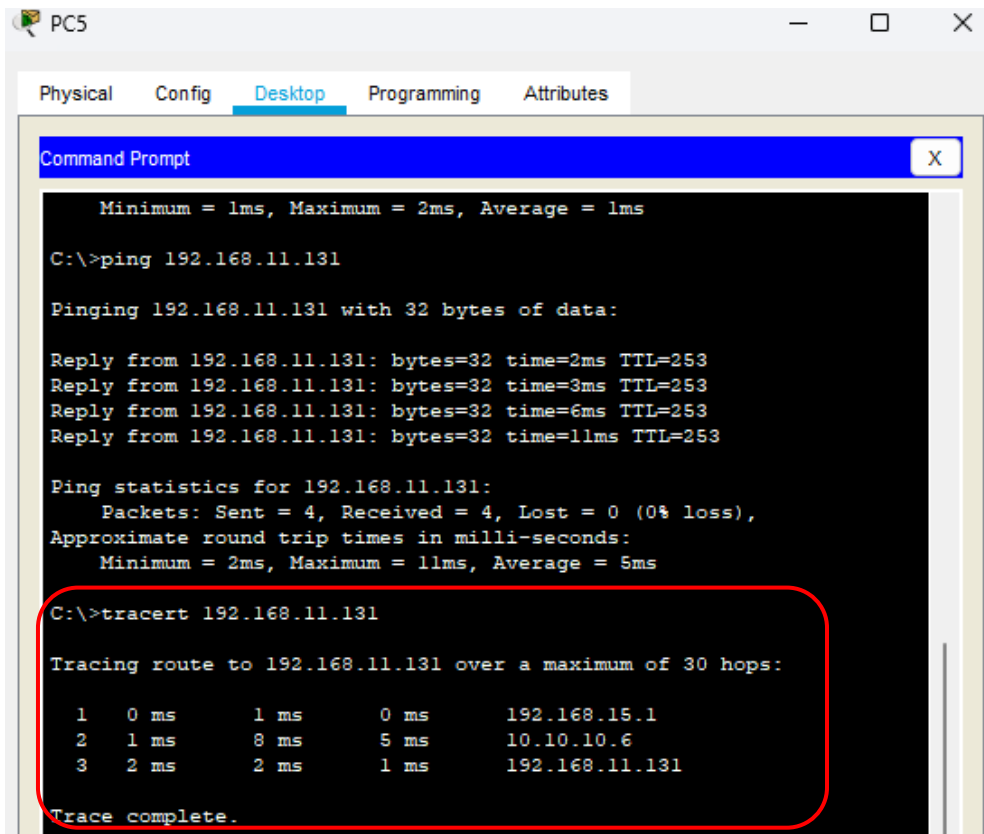
Pinging 192.168.11.131 with 32 bytes of data:

Reply from 192.168.11.131: bytes=32 time=2ms TTL=253
Reply from 192.168.11.131: bytes=32 time=3ms TTL=253
Reply from 192.168.11.131: bytes=32 time=6ms TTL=253
Reply from 192.168.11.131: bytes=32 time=11ms TTL=253

Ping statistics for 192.168.11.131:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 11ms, Average = 5ms
```

Only admin CCTV section can access CCTVs of all departments

So all department's CCTV's are centralized with the CCTV section of admin



The screenshot shows a Windows PC5 desktop with a Command Prompt window open. The window title is "Command Prompt". The desktop has tabs for "Physical", "Config", "Desktop", "Programming", and "Attributes". The Command Prompt shows the following output:

```
Minimum = 1ms, Maximum = 2ms, Average = 1ms

C:\>ping 192.168.11.131

Pinging 192.168.11.131 with 32 bytes of data:

Reply from 192.168.11.131: bytes=32 time=2ms TTL=253
Reply from 192.168.11.131: bytes=32 time=3ms TTL=253
Reply from 192.168.11.131: bytes=32 time=6ms TTL=253
Reply from 192.168.11.131: bytes=32 time=11ms TTL=253

Ping statistics for 192.168.11.131:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 11ms, Average = 5ms

C:\>tracert 192.168.11.131

Tracing route to 192.168.11.131 over a maximum of 30 hops:

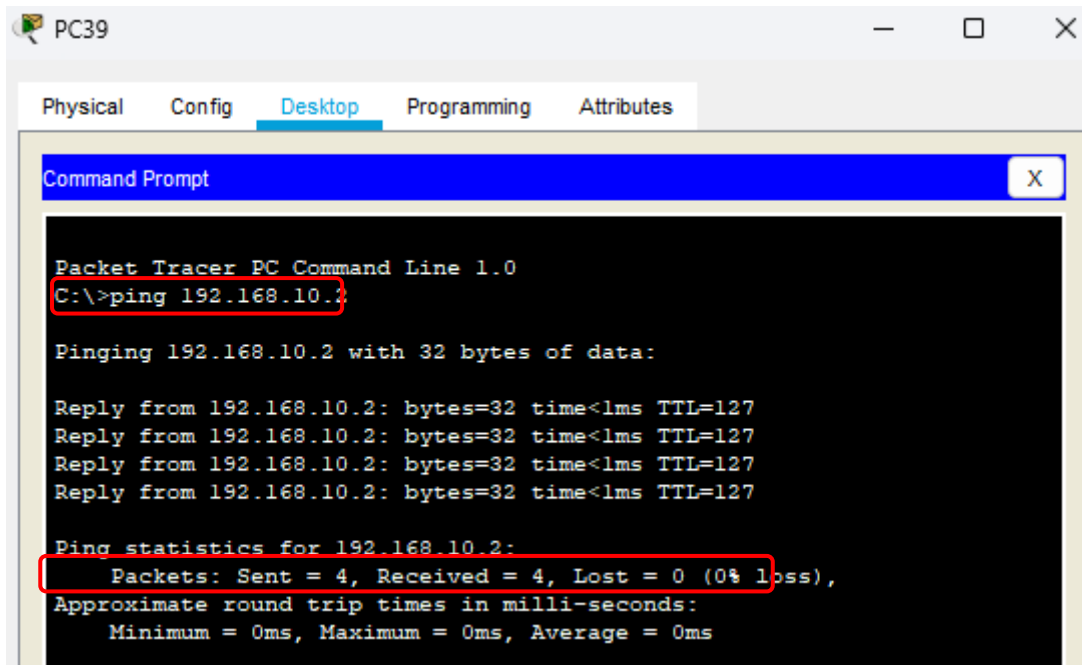
  1  0 ms      1 ms      0 ms      192.168.15.1
  2  1 ms      8 ms      5 ms      10.10.10.6
  3  2 ms      2 ms      1 ms      192.168.11.131

Trace complete.
```

4. 192.168.10.0/25 (Subnet Address of mech-staff)

192.168.10.1-192.168.10.126 (Range of mech-staff IP)

- Pinging from staff of another department to mech-staff



The screenshot shows a Packet Tracer PC window for PC39. The 'Desktop' tab is active, displaying a Command Prompt. The command 'C:\>ping 192.168.10.2' is entered and highlighted with a red box. The output shows four successful replies from 192.168.10.2 with 32 bytes of data, each taking less than 1ms and having a TTL of 127. The ping statistics for 192.168.10.2 are also highlighted with a red box, showing 4 packets sent, 4 received, and 0% loss.

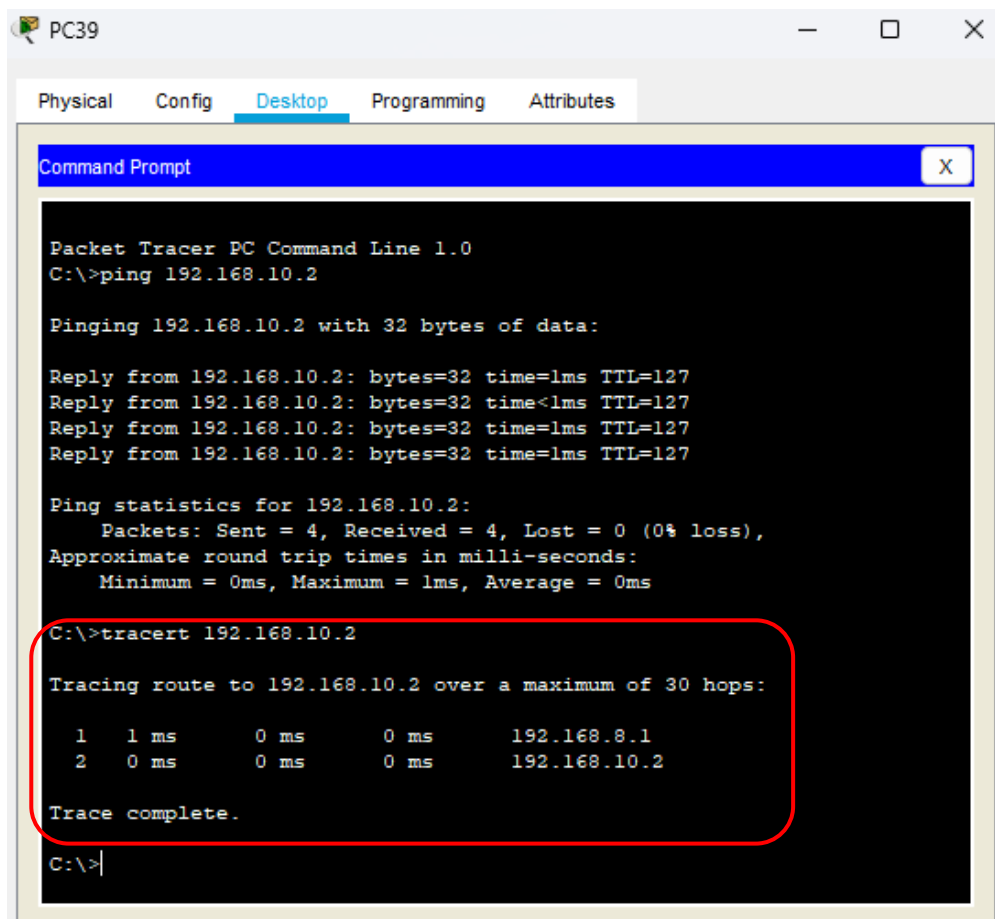
```
Packet Tracer PC Command Line 1.0
C:\>ping 192.168.10.2

Pinging 192.168.10.2 with 32 bytes of data:

Reply from 192.168.10.2: bytes=32 time<1ms TTL=127
Reply from 192.168.10.2: bytes=32 time<1ms TTL=127
Reply from 192.168.10.2: bytes=32 time<1ms TTL=127
Reply from 192.168.10.2: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.10.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

Each department's staff can communicate each other



The screenshot shows the same Packet Tracer PC window for PC39. The 'Desktop' tab is active, displaying a Command Prompt. The command 'C:\>ping 192.168.10.2' is entered and highlighted with a red box. The output shows four successful replies from 192.168.10.2 with 32 bytes of data, each taking less than 1ms and having a TTL of 127. The ping statistics for 192.168.10.2 are also highlighted with a red box, showing 4 packets sent, 4 received, and 0% loss. Below the ping output, the command 'C:\>tracert 192.168.10.2' is entered and highlighted with a red box. The output shows the tracing route to 192.168.10.2 over a maximum of 30 hops, with two hops shown: 1 ms to 192.168.8.1 and 0 ms to 192.168.10.2. The trace is complete.

```
Packet Tracer PC Command Line 1.0
C:\>ping 192.168.10.2

Pinging 192.168.10.2 with 32 bytes of data:

Reply from 192.168.10.2: bytes=32 time=1ms TTL=127
Reply from 192.168.10.2: bytes=32 time<1ms TTL=127
Reply from 192.168.10.2: bytes=32 time=1ms TTL=127
Reply from 192.168.10.2: bytes=32 time=1ms TTL=127

Ping statistics for 192.168.10.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>tracert 192.168.10.2

Tracing route to 192.168.10.2 over a maximum of 30 hops:

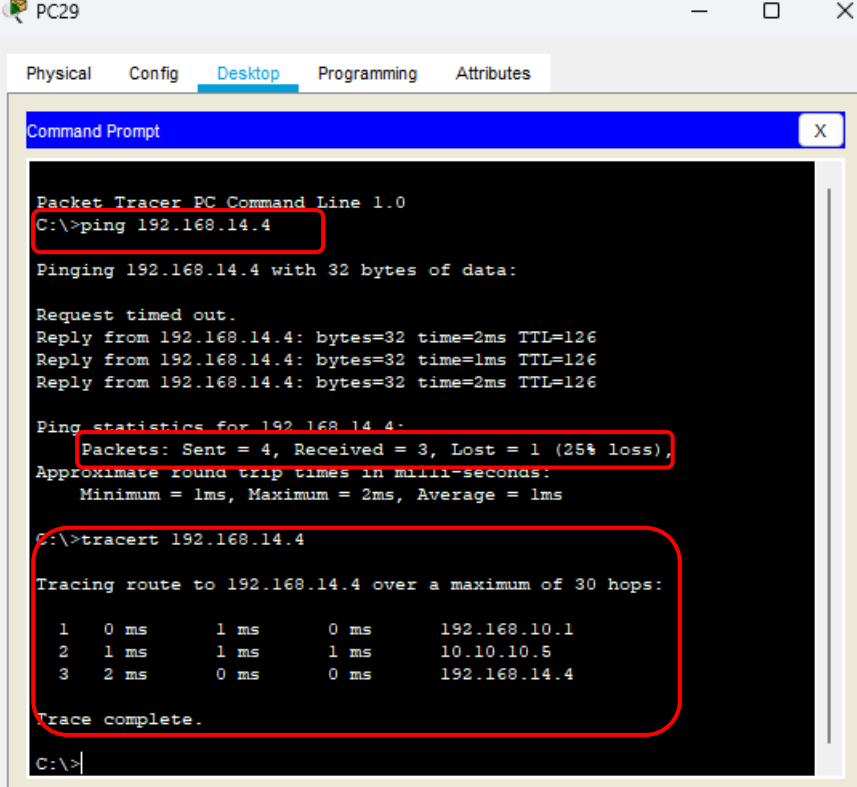
  1  1 ms    0 ms    0 ms    192.168.8.1
  2  0 ms    0 ms    0 ms    192.168.10.2

Trace complete.

C:\>
```

VLAN and routing functionality tests

- From mech-staff to admin-staff



PC29

Physical Config **Desktop** Programming Attributes

Command Prompt

```
Packet Tracer PC Command Line 1.0
C:\>ping 192.168.14.4

Pinging 192.168.14.4 with 32 bytes of data:

Request timed out.
Reply from 192.168.14.4: bytes=32 time=2ms TTL=126
Reply from 192.168.14.4: bytes=32 time=1ms TTL=126
Reply from 192.168.14.4: bytes=32 time=2ms TTL=126

Ping statistics for 192.168.14.4:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 2ms, Average = 1ms

C:\>tracert 192.168.14.4

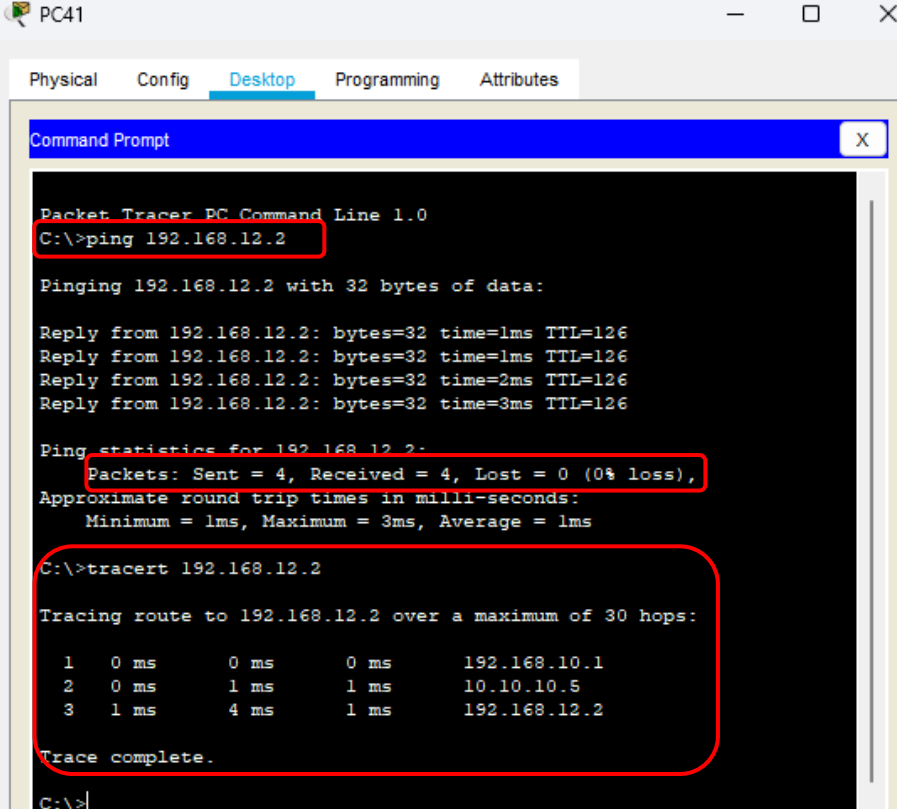
Tracing route to 192.168.14.4 over a maximum of 30 hops:

  0  0 ms    1 ms    0 ms    192.168.10.1
  1  1 ms    1 ms    1 ms    10.10.10.5
  2  2 ms    0 ms    0 ms    192.168.14.4

Trace complete.

C:\>
```

- From mech-staff to IDS-staff



PC41

Physical Config **Desktop** Programming Attributes

Command Prompt

```
Packet Tracer PC Command Line 1.0
C:\>ping 192.168.12.2

Pinging 192.168.12.2 with 32 bytes of data:

Reply from 192.168.12.2: bytes=32 time=1ms TTL=126
Reply from 192.168.12.2: bytes=32 time=1ms TTL=126
Reply from 192.168.12.2: bytes=32 time=2ms TTL=126
Reply from 192.168.12.2: bytes=32 time=3ms TTL=126

Ping statistics for 192.168.12.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 3ms, Average = 1ms

C:\>tracert 192.168.12.2

Tracing route to 192.168.12.2 over a maximum of 30 hops:

  0  0 ms    0 ms    0 ms    192.168.10.1
  1  0 ms    1 ms    1 ms    10.10.10.5
  2  1 ms    4 ms    1 ms    192.168.12.2

Trace complete.

C:\>
```

Scalability Validation

Department	Category	Subnet	Usable IPs
Computer	Staff	192.168.0.0/24	254
	Student	192.168.0.128/23	510
	Shared	192.168.2.0/25	126
	CCTV	192.168.3.128/25	126
EEE	Staff	192.168.4.0/24	254
	Student	192.168.5.0/23	510
	Shared	192.168.7.0/25	126
	CCTV	192.168.7.128/25	126
Civil	Staff	192.168.8.0/25	126
	Student	192.168.8.128/25	126
	Shared	192.168.9.0/25	126
	CCTV	192.168.9.128/25	126
Mechanical	Staff	192.168.10.0/25	126
	Student	192.168.10.128/25	126
	Shared	192.168.11.0/25	126
	CCTV	192.168.11.128/25	126
IDS	Staff	192.168.12.0/25	126
	Student	192.168.12.128/25	126
	Shared	192.168.13.0/25	126
	CCTV	192.168.13.128/25	126
Admin	Student	192.168.14.0/25	126
	Shared	192.168.14.128/25	126
	CCTV	192.168.15.0/25	126

Scalability in Networking

Scalability refers to the ability of a network to handle increasing amounts of traffic, users, or devices without compromising performance. In this case, the network is designed to accommodate up to **4,760 devices** across various VLANs (Virtual Local Area Networks).

Key aspects of scalability in this network:

- Subnetting Strategy:
The use of /24, /23, and /25 subnets ensures efficient IP allocation and allows for future expansion.
- VLAN Segmentation:
Different VLANs for Staff, Students, Shared resources, and CCTV reduce congestion and improve manageability.
- Future Growth:
If more devices are needed, additional VLANs or subnets can be created, ensuring long-term scalability.

Network Performance Considerations

Network performance is measured based on speed, reliability, and efficiency. Here are some factors affecting performance in this network:

1. Traffic Management:
 - VLANs prevent unnecessary traffic from spreading across the entire network.
 - Quality of Service (QoS) can be implemented to prioritize critical applications.
2. Bandwidth Allocation:
 - **CCTV traffic** may consume high bandwidth due to video streaming, potentially affecting other users.
 - **Student networks** might face congestion if many devices stream videos or transfer large files.
3. Latency and Congestion:
 - The network is designed to minimize latency by segregating traffic into VLANs.
 - High-bandwidth applications may require additional network upgrades in the future.
4. Monitoring and Optimization:
 - Implementing network monitoring tools can help track performance and detect congestion.
 - Future scalability can be improved by upgrading network infrastructure, such as switches and routers.

Detailed Explanation of Network Topology with Access Control

- ✓ Our network topology is designed for scalability, security, and efficient communication, ensuring that each department and user group has controlled access while maintaining network performance and traffic optimization.

1. VLAN-Based Segmentation and Inter-VLAN Communication

- ✓ To maintain security and reduce unnecessary traffic, the network is structured using Virtual LANs (VLANs). Each department has separate VLANs for staff, students, shared resources, and CCTV systems. This segmentation ensures that devices within a VLAN can communicate efficiently while restricting unnecessary interactions between different user groups.
- ✓ Inter-VLAN communication is enabled through Layer 3 Switching, which allows specific VLANs to communicate based on predefined rules while blocking unauthorized access. This setup reduces broadcast traffic, enhances security, and ensures optimized network performance.

2. Access Control Rules Implementation

I. Staff-Student Communication Restriction

- ✓ Each department's staff VLAN can access its own student VLAN, but not students from other departments. This restriction ensures that only authorized personnel can interact with students while preventing cross-departmental student access. This enhances departmental security and maintains organizational boundaries.

II. Restricted Access to Shared Resources

- ✓ Shared devices such as printers and file servers are placed in a dedicated shared VLAN within each department. Only the staff VLAN of the corresponding department is granted access to these resources, while students are strictly restricted. This prevents students from misusing or overloading shared resources and ensures controlled access for faculty and authorized personnel.

III. Centralized CCTV System for Admin Use

- ✓ Each department has a CCTV VLAN to monitor activities within the respective department. However, only the Admin CCTV VLAN has centralized access to all CCTV feeds across departments. Other users, including staff and students, do not have access to CCTV systems outside their own department. This centralized approach ensures high security and controlled surveillance management, allowing the administration to monitor all department CCTV feeds from a single control center.

IV. Staff Communication Across Departments

- ✓ To ensure smooth coordination between different departments, staff VLANs are allowed to communicate with one another. However, students are restricted from communicating with staff outside their department, preventing unnecessary interactions and maintaining structured communication channels. This approach allows faculty members from various departments to collaborate effectively while maintaining security policies for students.

3. Network Performance and Traffic Optimization

- ✓ The implementation of VLANs significantly reduces broadcast traffic, which enhances overall network performance. Since devices communicate only within their assigned VLANs, unnecessary network congestion is minimized. This is particularly beneficial for maintaining high-speed connectivity and efficient resource utilization.
- ✓ The Layer 3 Switch ensures faster inter-VLAN communication compared to traditional routers, leading to reduced latency and improved network performance. Additionally, Quality of Service policies are implemented to prioritize critical network traffic. For instance, CCTV feeds require higher bandwidth to ensure smooth video streaming, while shared resources and staff communication are prioritized over general student internet access. By managing traffic effectively, QoS helps prevent network slowdowns and optimizes data flow.

4. Scalability Considerations

- ✓ The network is designed with scalability in mind, allowing future expansion without disrupting existing functionality. New VLANs can be added seamlessly to accommodate additional departments, devices, or security policies. If more devices need to be connected in the future, additional subnets can be allocated while maintaining structured communication. The QoS policies and access control rules can also be updated as network demands increase, ensuring that the system remains efficient and secure over time.